Sequences, Series and Complex Numbers
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Abstract

Concepts regarding real numbers and real variable functions such as limits, continuity and convergences are introduced in real analysis. However, we can easily generalize the notions to functions with complex variables such as z. Throughout this project, we will compare different types of objects associated with the $\mathbb{R}$ and the complex plane $\mathbb{C}$, while highlighting similarities and interesting results between them.

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1 Introduction

1.1 Overview

The concepts of function and calculus should be familiar to us, with an approach of drawing graphs in order to illustrate the situation. For example, consider a function $f : \mathbb{R} \to \mathbb{R}$, then the process of differentiation, i.e. finding f', can be thought of finding the *slope* or *gradient* of the graph $y = f(x)$ on the (x, y) coordinate plane:

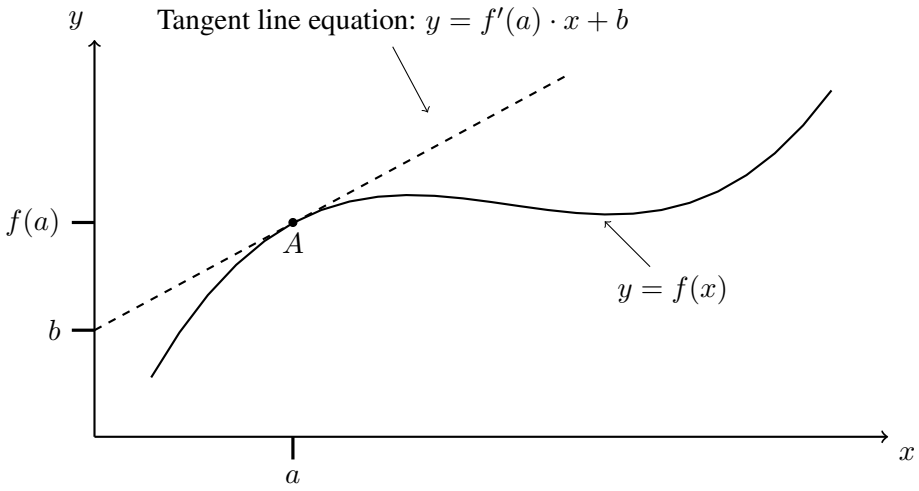


Figure 1: The idea of differentiation of a real variable function $f : \mathbb{R} \to \mathbb{R}$ is that it allows us to find the slope (hence the tangent line) of the function at a general point.

The concepts above are rigorously introduced, involving many definitions. For instance, the set of real numbers $\mathbb{R}$ is introduced to be the set extended from the rational numbers with the least upper bound property. The study of real variables and the properties of the set $\mathbb{R}$ is generally called **real analysis**. Main ideas of the subject in which the reader should be familiar with, include the ideas of continuity and differentiation, and this can be found in textbooks like [11, 19, 18].

It turns out that the rigorous approach we have can serve a better purpose than regurgitating knowledge that is terse. Also, it can *generalize* our discussions. Instead of looking at functions with real variables, the results from real analysis topics enables us to take a look at complex-valued objects. Such direction of study is another branch of mathematics called **complex analysis**, which will be the main focus of this project.

This paper aims to develop theories on complex numbers. We shall introduce functions f that map a complex number to another one, starting with the familiar topics of sequences, series and limits. This mapping is, in a sense, 'bigger' than a real-to-real mapping. We shall investigate ways to visualize common functions and regions. The theorems that we shall use is mainly taken from [14, Chapter 1 and 3].

Afterwards, we shall explore some applications of the topics that we developed, especially a view into geometry through stereographic projection. It is a concept of mapping every single point on a sphere onto a plane and hence, it first came

about historically as a problem in cartography, where the problem of mapping something that is spherical and in three dimensions to something that is flat and two dimensional. This allow us to see the earth on a paper, interested readers in geography may refer to [9] for more details.

1.2 Prerequisites

In this section, we shall recall some basic facts on complex numbers, as well as introducing basic notation which is going to be used throughout the report. More details can be found in [20, Chapter 1].

Definition 1.2.1. The set of complex numbers is defined as:

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\},$$

where the number $i = \sqrt{-1}$. Given a complex number $z = a + bi$, we shall denote the real part as $\mathrm{Re}(z) = a$ and the imaginary part as $\mathrm{Im}(z) = b$. We shall also define its complex conjugate $\overline{z} = a - bi$. Namely, given two complex numbers $z = a + bi$ and $w = c + di$:

$$z + w = (a + c) + (b + d)i, \quad zw = (a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

We can see that the sets $\mathbb{R}$ and $\mathbb{C}$ share a lot of common properties. For example:

Proposition 1.2.2.

1. $\mathbb{R}$ and $\mathbb{C}$ are both fields under the usual addition and multiplication operations. Hence, we can take the field axioms for granted. In particular, $\mathbb{R}$ is a subfield of $\mathbb{C}$.
2. The idea of distance can be used for elements of the sets $\mathbb{R}$ and $\mathbb{C}$. In particular, for real numbers, the distance is the absolute value $|a|$, and for complex numbers, the distance is the modulus $|z| = \sqrt{a^2 + b^2}$.

Proof. The proof for the above and the idea of fields is very elementary and is contained in most of the introductory textbooks about Real and Complex analysis, especially in [5, Section 1]. □

Thus, it is natural to extend various concepts in real analysis into complex analysis. However, complex numbers have one special feature that real numbers don't. For instance, any number $z = x + yi \in \mathbb{C}$ can be thought of as a vector $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$. Hence there is a function $\arg(z) = \theta$, where θ is the angle between $\begin{pmatrix} x \\ y \end{pmatrix}$ and the x-axis. This is illustrated in the Figure 2 shown below.

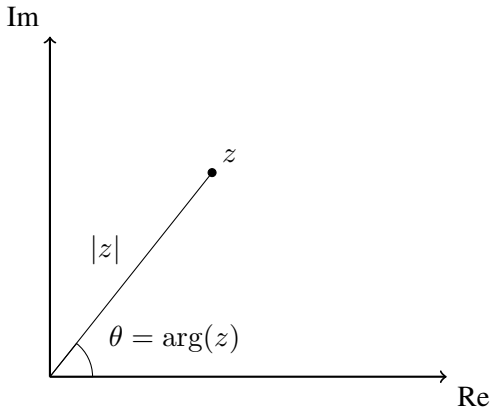


Figure 2: An Argand diagram of the complex number z. We can draw such diagrams because complex numbers can be represented by vectors in $\mathbb{R}^2$. Please note that the x-axis and y-axis are labelled as "Re" and "Im" respectively, meaning that the x-coordinate represents the real part of z and that the y-coordinate represents the imaginary part of z.

Also, we must avoid a slight caveat of the complex numbers:

Theorem 1.2.3. There is no strict ordering in $\mathbb{C}$. That is, there does **not** exist a relation $>$ such that for all $z, w \in \mathbb{C}$, the following three conditions holds at the same time:

1. $z, w > 0 \implies a + b > 0$ (Addition preserves order),
2. $z, w > 0 \implies ab > 0$ (Multiplication preserves order),
3. Exactly one of the three cases is true: $z > 0$, $z = 0$, $-z > 0$ (Trichotomy),

where $-z$ is the additive inverse of z.

Proof. We shall proceed to prove by contradiction. Firstly, we shall assume that the relation mentioned above exist. If $i > 0$, then by rule 2, taking $z = w = i$, we have that $i^2 = -1 > 0$. Now, since $-1 > 0$, and taking $z = i, w = -1$, we have that $(i)(-1) = -i > 0$, contradicting the trichotomy.
If, on the other hand, we have that $-i > 0$, then, by using rule 2 mentioned above, we can deduce that $(-i)(-i) = 1 > 0$. Now, taking $z = 1, w = i$ in rule 2 again, we have that $(1)(i) = i > 0$, once again contradicting the trichotomy.
We also know that $i = 0$ can't happen due to the uniqueness of an additive inverse in the field $\mathbb{C}$. Therefore, all possible cases of such an order have led to a contradiction. Therefore, such a relation cannot exist. □

As a result, we cannot do a direct term by term comparison when dealing with, for example, the sandwich test of convergences for sequences. But, with caution, we can break the argument down to two real variable sequences. This is introduced in the next section.

As later we shall investigate properties of some subsets of $\mathbb{C}$ and investigate the geometric significance of them, while quoting some basic results from set theory:

Theorem 1.2.4 (De Morgan's Law)**.** For an indexing set $\mathcal{I}$, firstly consider a family of sets $\{A_k : k \in \mathcal{I}\}$, and we have that

$$\left(\bigcup_{k\in\mathcal{I}} A_k\right)^c = \bigcap_{k\in\mathcal{I}} A_k^c, \left(\bigcap_{k\in\mathcal{I}} A_k\right)^c = \bigcup_{k\in\mathcal{I}} A_k^c,$$

where A^c denotes the complement of a set A. In particular, if $\mathcal{I} = \mathbb{N}$, then we will have that

$$\left(\bigcup_{k=1}^{\infty} A_k\right)^c = \bigcap_{k=1}^{\infty} A_k^c, \left(\bigcap_{k=1}^{\infty} A_k\right)^c = \bigcup_{k=1}^{\infty} A_k^c$$

Proof. It is an elementary proof, so it will not be included in here. Readers can refer to [10, Page 31] for details. □

2 Complex Sequences

Consider $\{x_n\}_{n\in\mathbb{N}}$, where for all $n \in \mathbb{N}$, $x_n \in \mathbb{R}$. From this we can also define a sequence with complex numbers:

$$\{z_n\}_{n\in\mathbb{N}} = \{z_1, z_2, ...\},$$

where $z_n \in \mathbb{C}$ for all n.

2.1 Properties of sequences

As mentioned in Section 1, knowledge of usual arithmetic of sequence is expected from the reader, and from hence it should be obvious that we can decompose a sequence into their real parts and imaginary parts. That is, we can deduce that:

Definition 2.1.1. Any complex sequence, denoted as $\{z_n\}_{n\in\mathbb{N}}$, is of the following form:

$$\{z_n\}_{n\in\mathbb{N}} = \{x_n\}_{n\in\mathbb{N}} + i\{y_n\}_{n\in\mathbb{N}} \tag{2.1}$$

where $\{x_n\}_{n\in\mathbb{N}}, \{y_n\}_{n\in\mathbb{N}}$ are real sequences.

The discussion of real-valued sequences can be found in [2, Chapter 3]. The decomposition of complex sequences into real-valued sequences implies that much of the theory developed for real sequences can be carried over. In particular:

Definition 2.1.2. The **limit** of a complex sequence $\{z_n\}_{n\in\mathbb{N}}$ is a complex number z, in which it is denoted as $\lim_{n\to\infty} z_n = z$, such that for all $\varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that

$$n > N \quad \text{implies} \quad |z_n - z| < \varepsilon$$

If such z exists, we say that $\{z_n\}_{n\in\mathbb{N}}$ is **convergent** or **converges to** z.

Definition 2.1.3. A sequence $\{z_n\}_{n\in\mathbb{N}}$ is **bounded** if and only if there exists an $M \in \mathbb{R}$ such that $|z_n| < M$ for all $n \in \mathbb{N}$.

Theorem 2.1.4. Supposing $\{z_n\}_{n\in\mathbb{N}}$ is a sequence of complex numbers and that $n \in \mathbb{N}$, the following two statements are equivalent:

1. $\lim_{n\to\infty} z_n = z = x + iy$,
2. $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$,

Proof. Firstly, we will assume that (2.) holds, that is, assume that $\lim_{n\to\infty} x_n = x$ and that $\lim_{n\to\infty} y_n = y$ holds. Let $\varepsilon > 0$, from the definition of the limit of a real sequence. Therefore, there exists $N_1 \in \mathbb{N}$ such that for $n > N_1$, $|x_n - x| < \dfrac{\varepsilon}{2}$ and $N_2 \in \mathbb{N}$ such that for $n > N_2$, $|y_n - y| < \dfrac{\varepsilon}{2}$. Let $N = \max(N_1, N_2)$. By the triangle inequality,

$$|z_n - z| \leqslant |x_n - x| + |y_n - y| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \qquad , n > N.$$

Thus, by definition, $\lim_{n\to\infty} z_n = z$.

Conversely, if (1.) holds, then $\lim_{n\to\infty} z_n = z$. Then, for some $n > N, N \in \mathbb{N}$, $|z_n - z| \leqslant \varepsilon$. Let $x = \text{Re}z$. Then we have that:

$$|x_n - x| = \text{Re}|z_n - z| \leqslant |z_n - z| \leqslant \varepsilon.$$

Thus, $\lim_{n\to\infty} x_n = x$. A similar argument follows for y_n by taking the imaginary part of z_n instead, proving the statements are equivalent. □

It is important for us to understand how these complex sequences converge by looking at a graphical interpretation of what these definitions mean for the complex plane. As the complex numbers lie on a plane and not a line, ε is thought of as a 'ball' around the convergence point from which $N \in \mathbb{N}$ great enough such that z_n never escapes. The formal definition of it, which is called an ε-neighbourhood around z_0 will be mentioned in Definition (3.2.1). See Figure 3 which illustrates this convergence on the complex plane.

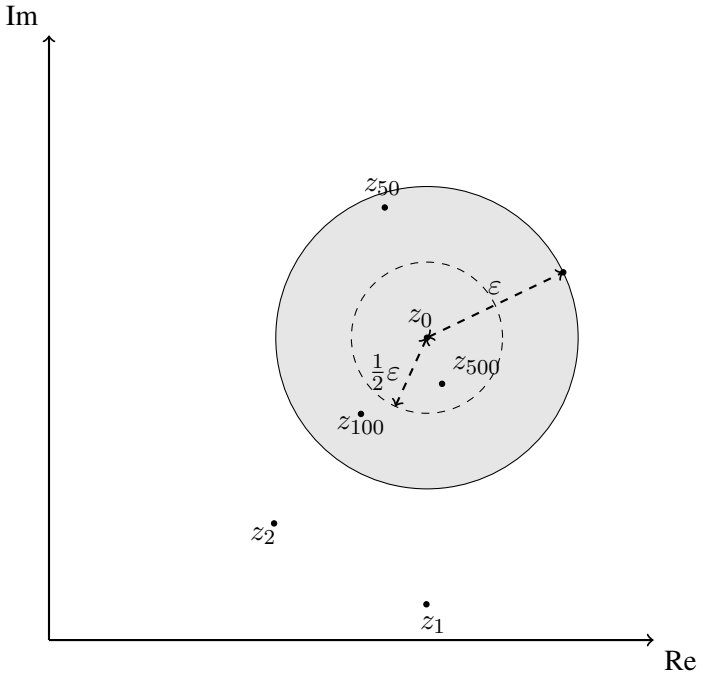


Figure 3: A graphical representation of an arbitrary sequence that converges to z_0. We can see that all terms of the sequence $\{z_n\}_{n\in\mathbb{N}}$ will eventually be inside an ε-neighbourhood of z_0 if n is sufficiently large. This works for any $\varepsilon > 0$ (hence $\dfrac{1}{2}\varepsilon$ works too).

Below, we shall provide more examples regarding complex sequences.

Example. Let $z_n = \left(\dfrac{1}{1+i}\right)^n$,$n \in \mathbb{N}$, and we claim that $\lim_{n\to\infty} z_n = 0$.

Proof. Let $\varepsilon > \frac{1}{n}$.Then,

$$|z_n - 0| = \left|\left(\frac{1}{1+i}\right)^n\right| = \left|\frac{1}{1+i}\right|^n = \frac{1}{|1+i|^n} = \frac{1}{\sqrt{2}^n} = \frac{1}{2^{\frac{n}{2}}} \leqslant \frac{1}{n} \leqslant \varepsilon.$$

Thus, by definition, $\lim_{n\to\infty} z_n = 0$. □

Example. Let $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and define a sequence $\{z_n\}_{n\in\mathbb{N}}$ with $z_n = \omega^n$. We will now show that $\{z_n\}_{n\in\mathbb{N}}$ is not convergent.

Note that, by direct calculation, $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \omega^3 = 1$, then $\omega^4 = \omega$, $\omega^5 = \omega^2$, et cetera. So the sequence z_n only ever takes three values, or:

$$\{z_n\}_{n\in\mathbb{N}} = \{\omega, \omega^2, 1, \omega, \omega^2, 1, ...\}$$

Hence by Theorem 2.1.4, it suffices to show that the real part of z_n doesn't converge. Indeed, if we extract the real part sequence according to (2.1) we have:

$$\{x_n\}_{n\in\mathbb{N}} = \left\{-\frac{1}{2}, -\frac{1}{2}, 1, -\frac{1}{2}, -\frac{1}{2}, 1, ...\right\}$$

which obviously doesn't converge, as required.

2.2 Cauchy sequences

Just like its real counterpart, the definition for a complex Cauchy sequence is almost identical to the definition in real analysis.

Definition 2.2.1. A sequence of complex numbers $\{z_n\}_{n\in\mathbb{N}}$ is a **Cauchy Sequence** of complex numbers if, for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that if $m, n > N$ then $|z_m - z_n| < \varepsilon$.

Lemma 2.2.2. Every Cauchy sequence of complex numbers is bounded.

Proof. Let $\{z_n\}_{n\in\mathbb{N}}$ be a complex-valued sequence. By definition, there exists $N \in \mathbb{N}$ such that for $m, n > N$, $|z_m - z_n| < 1$. By setting $m = N + 1 > N$ we can also say $|z_n| = |z_n - z_m + z_m|$. Thus, by the triangle inequality,

$$|z_n| \leqslant |z_n - z_m| + |z_m| < 1 + |z_m| = 1 + |z_{N+1}|.$$

Therefore for $m = N+1$, z_n is bounded while $n > N$. By taking $M \in \mathbb{C}$ to be equal to $\max(|z_1|, |z_2|, ..., |z_N|, 1 + |z_{N+1}|)$. Therefore, the fact $|z_n| < M$ for all $n \in \mathbb{N}$ so z_n is a bounded sequence must hold. Hence, all Cauchy sequences are bounded. □

Theorem 2.2.3. (Cauchy's Criterion for Convergence) An infinite sequence of complex numbers is convergent if and only if it is a Cauchy sequence.

Proof. Firstly, we shall prove that all complex convergent sequences are Cauchy. Assume $\{z_n\}_{n\in\mathbb{N}}$ is a convergent complex sequence with $\lim_{n\to\infty} z_n = z$. Let $\varepsilon > 0$, then there must exist some $N \in \mathbb{N}$ such that for $m, n > N$, $|z_n - z| < \frac{\varepsilon}{2}$ and also $|z_m - z| < \frac{\varepsilon}{2}$. Hence, by using the triangle inequality,

$$|z_n - z_m| = |(z_n - z) + (z - z_m)| \leqslant |z_n - z| + |z_m - z| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus, any convergent sequence is a Cauchy Sequence.

Conversely, we shall show how any complex Cauchy sequence $\{z_n\}_{n\in\mathbb{N}}$ is convergent. By Lemma 2.2.2, all Cauchy sequences are bounded. By letting $z_n = x_n + iy_n$, we can see that we can use the Bolzano-Weierstrass Theorem to say

that z_n contains a convergent sub-sequence, $\{z_{n_k}\}_{k\in\mathbb{N}}$. Also, we let $\lim_{k\to\infty} z_{n_k} = z$, and claim that z_n also converges to this point z. Let $\varepsilon > 0$, since z_n is a Cauchy sequence there exists $N \in \mathbb{N}$ such that for $n, m > N$, $|z_m - z_n| < \dfrac{\varepsilon}{2}$. Let k be large enough so that $n_k > N$. It follows that, by using the triangle inequality,

$$|z_n - z| \leqslant |z_n - z_{n_k}| + |z_{n_k} - z| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus, the fact $\lim_{n\to\infty} z_n = z$ holds, as required. □

Example. We claim that the complex sequence $\{z_n\}_{n\in\mathbb{N}}$, where $z_n = \dfrac{in^2 + 3n}{10n^2 + in + 10i}$, is a Cauchy sequence. By proving that z_n is convergent, z_n must also be a Cauchy sequence. It follows that:

$$\lim_{n\to\infty}\left(\frac{in^2 + 3n}{10n^2 + in + 10i}\right) = \lim_{n\to\infty}\left(\frac{i + \dfrac{3}{n}}{10 + \dfrac{i}{n} + \dfrac{10i}{n^2}}\right).$$

By using the standard result that $\lim_{n\to\infty}\left(\dfrac{1}{n^\alpha}\right) = 0$ where $\alpha \geqslant 1$ and also the fact that $\left(10 + \dfrac{i}{n} + \dfrac{10i}{n^2}\right) \neq 0$, we can compute the limit of z_n by the following:

$$\lim_{n\to\infty}\left(\frac{i + \dfrac{3}{n}}{10 + \dfrac{i}{n} + \dfrac{10i}{n^2}}\right) = \frac{i + 0}{10 + 0 + 10i \times 0} = \frac{i}{10},$$

hence z_n converges to $\dfrac{i}{10}$ also making it a Cauchy sequence, as required.

Indeed we can see that most of the properties of complex sequences is extremely similar to that of the real ones.

3 Limits of a Complex Function

3.1 Complex Functions

Before considering complex functions and their limits, we shall recall some basic facts and properties of limits of real valued functions that we covered in real analysis without proof, and refer back to these later on.

Theorem 3.1.1. Let $X \subseteq \mathbb{R}$, $f, g : X \to \mathbb{R}$, let x_0 be an accumulation point of X and $a, b \in \mathbb{R}$ such that $\lim_{x\to x_0} f(x) = a$, $\lim_{x\to x_0} g(x) = b$ then we have that,

1. $$\lim_{x\to x_0} (f(x) + g(x)) = a + b,$$
2. $$\lim_{x\to x_0} (f(x)g(x)) = ab,$$
3. if $b \neq 0$ then,
$$\lim_{x\to x_0} \left(\frac{f(x)}{g(x)}\right) = \frac{a}{b}.$$

Corollary 3.1.2. From (1) and (2) in Theorem 3.1.1, we can clearly show by induction that, by letting $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, be any polynomial, where $a_n, \ldots, a_0 \in \mathbb{R}$, then with $x_0 \in \mathbb{R}$,

$$\lim_{x\to x_0} P(x) = P(x_0).$$

A complex function is a function where the domain and codomain are subsets of $\mathbb{C}$, that is, let X, Y be non-empty subsets of $\mathbb{C}$, then a complex function f defined on X is a map that assigns each $z \in X$ a value $w \in Y$. We shall write $f : X \to Y, f(z) = w$ to denote this. We know that, for $x, y, u, v \in \mathbb{R}$, we can write the following:

$$f(x + iy) = f(z) = w = u + iv.$$

Hence, we can equate the real and complex parts of the left and right hand side, $u = \text{Re}(f(x + iy)), v = \text{Im}(f(x + iy))$ to clearly show that the real and complex part of $f(z)$ are functions of the real numbers x, y associated with z. Therefore we can write the following expression:

$$f(x + iy) = u(x, y) + iv(x, y),$$

where $u, v : U, V \subseteq \mathbb{R}^2 \to \mathbb{R}$ and that u, v are determined by the function f.

Example. Let $X = \{w \in \mathbb{C} : |w| < 1\}, f : X \to \mathbb{C}, f(z) = z^3$. Then by letting $z = x + iy$ for $x, y \in \mathbb{R}$, we have that:

$$\begin{aligned} f(z) &= f(x + iy) \\ &= (x + iy)^3 \\ &= x^3 + 3ix^2y - 3xy^2 - iy^3 \\ &= (x^3 - 3xy^2) + i(3x^2y - y^3) \\ &= u(x, y) + iv(x, y), \end{aligned}$$

where $u(x, y) = x^3 - 3xy^2$ and $v(x, y) = 3x^2y - y^3$.

Now, we shall extend the idea of a limit into the complex plane, and this will allow us to define continuity and differentials in the complex setting. To do this we shall firstly introduce the idea of a neighbourhood.

3.2 Neighbourhoods

We have seen that a complex number z can be defined by two real numbers x and y. Hence, the complex field can be viewed as the set $\mathbb{R}^2$ with the operation of complex multiplication. Therefore, we can think of z as a vector in $\mathbb{R}^2$ and we can define the modulus of a complex number as the norm of the corresponding vector in $\mathbb{R}^2$, i.e. $|z| = |x + iy| = \|(x, y)\| = \sqrt{x^2 + y^2}$. We will use this to define a 'neighbourhood'.

Definition 3.2.1. For any finite $z \in \mathbb{C}$ and $r \in \mathbb{R}, r > 0$, the **r-neighbourhood around** z is defined as of the following:

$$D(z : r) := \{w \in \mathbb{C} : |w - z| < r\}.$$

Also, the **deleted r-neighbourhood around** z is defined as of the following:

$$D(z : r)\backslash\{z\} := \{w \in \mathbb{C} : 0 < |w - z| < r\}.$$

Note that it will only make sense to discuss a neighbourhood around a point. Geometrically speaking, the r-neighbourhood around z_0 consists of all the points inside the circle centered at z_0 with a radius r, excluding the boundary. While the deleted r-neighbourhood around z_0 is just the r-neighbourhood around z_0, with the point z_0 excluded. Figure 4 and Figure 5 illustrate the above sets.

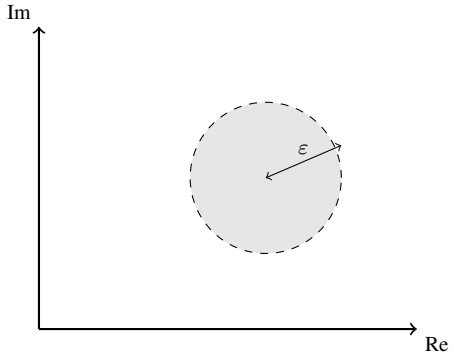


Figure 4: An r-neighbourhood.

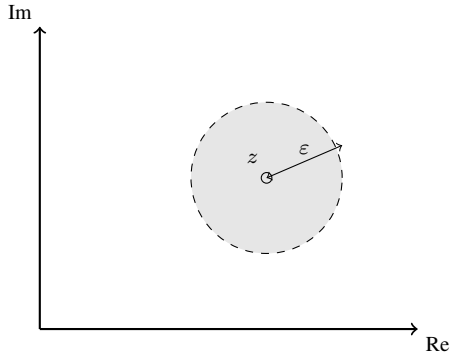


Figure 5: A deleted r-neighbourhood.

We have, from real analysis, the idea of an accumulation point which is used in the definition of a limit. Here we will formally define this in the general setting and then we will look at an example in the complex plane.

Definition 3.2.2. An **accumulation point** of a set is a point in which is the limit of a sequence of points in the set that is distinct from the point itself. Formally, let X be a set, then x is an accumulation point of the set X if and only if there exists a sequence $\{x_n\}_{n\in\mathbb{N}}$, with $x_n \in X, x_n \neq x$ for all $n \in N$ such that $\lim_{n\to\infty} x_n = x$.

Example. Let a complex function f be defined on a deleted disk of radius 1 centered at the origin, with the condition that 0 is not in the domain. We can show that, although 0 is not in the domain, it is in fact an accumulation point, since the sequence $\left\{\frac{1}{n}\right\}_{n\in\mathbb{N}}$ satisfies the following condition: $\left|\frac{1}{n}\right| < 1, \frac{1}{n} \neq 0$, for all $n \in N$ and $\lim_{n\to\infty}\left(\frac{1}{n}\right) = 0$.
Because of this, although f is not defined at 0, we can look at the limit as f approaches 0.

We can now use these ideas to define the limit of a complex function at a point z_0.

Definition 3.2.3. Let $X \subseteq \mathbb{C}$ such that $X \neq \emptyset$, $f : X \to \mathbb{C}$ and let $z_0 \in X$ be an accumulation point of X. Thus we can say that f has a **limit** $w \in \mathbb{C}$ as z tends to z_0 if and only if $f(z)$ can be made arbitrarily close to w by making z close but not equal to z_0. Formally, we can define the following:

$$\lim_{z \to z_0} f(z) = w$$

if and only if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that:

$$z \in D(z_0 : \delta) \backslash \{z_0\} \implies f(z) \in D(w : \varepsilon).$$

Example. Let $X = \{w \in \mathbb{C} : |w| < 1\}$ and let $f : X \to \mathbb{C}$ be a function such that $f(z) = \dfrac{\overline{z}}{2i}$. We can claim that:

$$\lim_{z \to i} f(z) = \frac{-1}{2},$$

since, by letting $\varepsilon > 0$ be given, we can deduce that $\delta = 2\varepsilon$. Therefore, we have the following equation stated below.

$$\begin{aligned} z \in D(i, 2\varepsilon) \backslash \{i\} &\implies 0 < |z - i| < 2\varepsilon \\ &\implies \frac{|x + iy - i|}{2} < \varepsilon \\ &\implies \left| \frac{x}{2} - \frac{iy}{2} + \frac{i}{2} \right| < \varepsilon \\ &\implies \left| \frac{\overline{z}}{2} - \frac{-1}{2} \right| < \varepsilon \\ &\implies f(z) \in D\left(-\frac{1}{2} : \varepsilon \right). \end{aligned}$$

Please note that this is true for all $\varepsilon > 0$, and therefore,

$$\lim_{z \to i} f(z) = \frac{-1}{2}$$

.

As we have seen, a complex function can be interpreted as a sum of two functions with two variables, and thus we can determine its real and imaginary parts. Therefore, we can consider the limits of these functions in order to determine the limit of the function itself. However, in order to do this, we must first define the double limit, which is fairly analogous to the limit.

Definition 3.2.4. Let $X \subseteq \mathbb{R}^2$ be defined. Also, let $f : X \to \mathbb{R}$ be a function and let $(x_0, y_0) \in \mathbb{R}^2$. Then we can say that the function f has a **double limit** $a \in \mathbb{R}$ as (x, y) tends to (x_0, y_0) if and only if $f(x, y)$ can be made arbitrarily close to a by making x close to x_0 and y close but not equal to y_0. Formally we can say the following:

$$\lim_{(x,y) \to (x_0, y_0)} f(x, y) = a$$

if and only if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$x \in D(x_0 : \delta) \backslash \{x_0\}, y \in D(y_0 : \delta) \backslash \{y_0\} \implies f(x, y) \in D(a : \varepsilon).$$

We can now use this in the next theorem, which is of the following:

Theorem 3.2.5. Let $X \subseteq \mathbb{C}$ and let $f : X \to \mathbb{C}, f(z) = f(x+iy) = u(x,y) + iv(x,y)$ where $x, y \in \mathbb{R}, u, v : U, V \subseteq \mathbb{R}^2 \to \mathbb{R}$. Suppose that $z_0 = x_0 + iy_0$ and that $w_0 = u_0 + iv_0$. Therefore we can deduce the following:

1. $$\lim_{z \to z_0} f(z) = w_0$$

   if and only if

2. $$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0, \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0.$$

Proof. We will prove this equivalency by showing that (2) $\implies$ (1) and that (1) $\implies$ (2). So, assuming (2) to be true, we have that,

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0.$$

Therefore, letting an $\varepsilon > 0$ be given, there must exist a $\delta_1 > 0$ such that:

$$x \in D(x_0 : \delta_1)\backslash\{x_0\}, y \in D(y_0 : \delta_1)\backslash\{y_0\} \implies u(x,y) \in D\left(u_0 : \frac{\varepsilon}{2}\right).$$

From this, we have that:

$$\begin{aligned} z \in D(z_0 : \delta_1)\backslash\{w_0\} &\implies 0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta_1 \\ &\implies 0 < |x - x_0|, |y - y_0| < \delta_1 \\ &\implies x \in D(x_0 : \delta_1)\backslash\{x_0\}, y \in D(y_0 : \delta_1)\backslash\{y_0\} \\ &\implies u(x,y) \in D\left(u_0 : \frac{\varepsilon}{2}\right) \end{aligned}$$

We also have from (2) that:

$$\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0,$$

and, from using similar reasoning, we can show that there must exist $\delta_2 > 0$ such that:

$$z \in D(z_0 : \delta_2)\backslash\{w_0\} \implies v(x,y) \in D\left(v_0 : \frac{\varepsilon}{2}\right).$$

Let $\delta < \min\{\delta_1, \delta_2\}$. Therefore we have that:

$$\begin{aligned} z \in D(w_0 : \delta)\backslash\{w_0\} &\implies u(x,y) \in D\left(u_0 : \frac{\varepsilon}{2}\right), v(x,y) \in D\left(v_0 : \frac{\varepsilon}{2}\right) \\ &\implies |v(x,y) - v_0|, |u(x,y) - u_0| < \frac{\varepsilon}{2} \\ &\implies |u(x,y) - u_0 + i(v(x,y) - v_0)| \leqslant |v(x,y) - v_0| + |u(x,y) - u_0| < \varepsilon \\ &\implies |u(x,y) + iv(x,y) - (u_0 + v_0)| < \varepsilon \\ &\implies |f(x,y) - z_0| < \varepsilon \\ &\implies f(x,y) \in D(z_0 : \varepsilon), \end{aligned}$$

and thus,

$$\lim_{z\to z_0} f(z) = w_0.$$

Now we suppose (1) holds, then we can deduce that:

$$\lim_{z\to z_0} f(z) = w_0.$$

So letting $\varepsilon > 0$ be given, there must exist a $\delta > 0$ such that:

$$z \in D(z_0, \sqrt{2}\delta)\backslash\{w_0\} \implies f(z) \in D(w_0 : \varepsilon).$$

We then have the fact that:

$$\begin{aligned}
x \in D(x_0 : \delta)\backslash\{x_0\}, y \in D(y_0 : \delta)\backslash\{y_0\} &\implies 0 < |x - x_0|, |y - y_0| < \delta_1 \\
&\implies 0 < |x - x_0 + i(y - y_0)| < \sqrt{2}\delta \\
&\implies 0 < |x + iy - (x_0 - iy_0)| < \sqrt{2}\delta \\
&\implies 0 < |z - z_0| < \sqrt{2}\delta \\
&\implies z \in D(z_0, \sqrt{2}\delta)\backslash\{w_0\} \\
&\implies f(z) \in D(w_0 : \varepsilon) \\
&\implies |f(z) - w_0| < \varepsilon \\
&\implies |u(x,y) + iv(x,y) - (u_0 + iv_0)| < \varepsilon \\
&\implies |u(x,y) - u_0 + i(v(x,y) - v_0)| < \varepsilon \\
&\implies |u(x,y) - u_0|, |v(x,y) - v_0| < \varepsilon \\
&\implies u(x,y) \in D(u_0 : \varepsilon), v(x,y) \in D(v_0 : \varepsilon),
\end{aligned}$$

and so,

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0, \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

holds, thus the proof is complete. □

Example. We will use Example 3.1 and the fact that $X = \{w \in \mathbb{C} : |w| < 1\}$ and that $f : X \to \mathbb{C}, f(x) = x^3$. Therefore, we can show that, with z_0 being $x_0 + iy_0$ which is equal to $\frac{-1}{2} + i\frac{\sqrt{3}}{2}$. Thus we can deduce that $\lim_{z\to z_0} f(z) = 1$. Therefore, we have shown that

$$f(z) = f(x + iy) = u(x,y) + iv(x,y),$$

where, $u(x,y) = x^3 - 3xy^2, v(x,y) = y^3 - 3x^2y$. We also have that:

$$\begin{aligned}
\lim_{(x,y)\to(x_0,y_0)} u(x,y) &= \lim_{(x,y)\to(x_0,y_0)} \left(x^3 - 3xy^2\right) \\
&= \lim_{(x,y)\to(x_0,y_0)} \left(x^3\right) - 3 \lim_{(x,y)\to(x_0,y_0)} \left(xy^2\right) \\
&= \lim_{x\to\frac{-1}{2}} \left(x^3\right) - 3 \lim_{x\to\frac{-1}{2}} (x) \lim_{y\to\frac{\sqrt{3}}{2}} \left(y^2\right) \\
&= \left(\frac{-1}{2}\right)^3 - 3\left(\frac{-1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)^2 \quad \text{(by Corollary (3.1.2))} \\
&= \frac{-1}{8} + \frac{3 \cdot 1 \cdot 3}{2 \cdot 4} \\
&= 1.
\end{aligned}$$

Therefore, it is easy to see that:

$$\begin{aligned}\lim_{(x,y)\to z_0} v(x,y) &= \lim_{(x,y)\to z_0} [y^3 - 3x^2y] \\ &= \left(\frac{\sqrt{3}}{2}\right)^3 - 3\left(\frac{-1}{2}\right)^2\left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{3\sqrt{3}}{8} - \frac{3\cdot 1\cdot\sqrt{3}}{4\cdot 2} \\ &= 0,\end{aligned}$$

and hence, by Theorem (3.2.5), we have that:

$$\begin{aligned}\lim_{z\to z_0} f(z) &= \lim_{(x,y)\to z_0} f(x+iy) \\ &= \lim_{(x,y)\to z_0} u(x,y) + i \lim_{(x,y)\to z_0} v(x,y) \\ &= 1 + i\cdot 0 \\ &= 1.\end{aligned}$$

3.3 Limits at infinity

Please note that our definition of a neighbourhood doesn't include the point at infinity. Therefore in order to consider limits at infinity, we must extend our definition of a neighbourhood to include the neighbourhood around infinity. Looking at limits at infinity of real functions is fairly rudimental since we can order the real numbers, giving us two points at infinity that we can approach. However, since we cannot order the complex numbers, there are infinitely many points that we can associate with infinity, and this makes limits involving infinity harder to define in the complex setting. To talk about infinite points, we need to rely on the idea of the stereographic projection which will be covered in depth later on in Section (6.1). But for now, we will take the definition for granted.

Definition 3.3.1. For any $r \in \mathbb{R}, r > 0$, the **r-neighbourhood around infinity** is defined as:

$$D(\infty : r) := \left\{ w \in \mathbb{C} : \frac{1}{r} < |w| \right\},$$

and similarly, the **deleted r-neighbourhood around infinity** is defined as:

$$D(\infty : r)\backslash\{\infty\} := \left\{ w \in \mathbb{C} : \frac{1}{r} < |w| < \infty \right\}.$$

With the neighbourhood around infinity now defined, we can therefore justify that:

$$\lim_{z\to z_0} f(z) = w_0$$

is now defined when z_0, w_0 or both are infinite, and the definitions follow by replacing the neighbourhood around the relevant point, with the neighbourhood being around infinity. Therefore we have that:

$$\lim_{z\to\infty} f(z) = w_0$$

if and only if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that,

$$z \in D(\infty : \delta)\backslash\{\infty\} \implies f(z) \in D(w_0 : \varepsilon).$$

We shall now prove the following theorem to make limits involving infinity slightly easier to work with.

Theorem 3.3.2. Let $f : X \subseteq \mathbb{C} \to \mathbb{C}, z_0, w_0 \in \mathbb{C}$ be a function such that z_0 is an accumulation point of X. Then we can deduce that:

1. $$\lim_{z\to z_0} f(z) = \infty \iff \lim_{z\to z_0} \frac{1}{f(z)} = 0,$$
2. $$\lim_{z\to\infty} f(z) = w_0 \iff \lim_{z\to 0} f\left(\frac{1}{z}\right) = w_0,$$
3. $$\lim_{z\to\infty} f(z) = \infty \iff \lim_{z\to 0} \frac{1}{f(\frac{1}{z})} = 0.$$

Proof. We will just prove (3) as the proofs of (1) and (2) follow on from (3). We have that (3) is true if and only if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that if $z \in D(\infty : \delta)\backslash\{\infty\}$, then $f(z) \in D(\infty : \varepsilon)$, and this is equivalent to the following:

$$\begin{aligned}
\frac{1}{\delta} < |z| < \infty \implies \frac{1}{\varepsilon} < |f(z)| &\iff \frac{1}{\delta} < \left|\frac{1}{z}\right| < \infty \implies \frac{1}{\varepsilon} < \left|f\left(\frac{1}{z}\right)\right| \\
&\iff 0 < |z| < \delta \implies \left|\frac{1}{f\left(\frac{1}{z}\right)}\right| < \varepsilon \\
&\iff 0 < |z - 0| < \delta \implies \left|\frac{1}{f\left(\frac{1}{z}\right)} - 0\right| < \varepsilon \\
&\iff z \in D(0, \delta)\backslash\{0\} \implies \frac{1}{f\left(\frac{1}{z}\right)} \in D(0, \varepsilon) \\
&\iff \lim_{z\to 0} \frac{1}{f\left(\frac{1}{z}\right)} = 0.
\end{aligned}$$

□

Now we shall look at some of the basic properties of complex limits.

3.4 Properties of Limits

From Theorem (3.2.5), we know that the limit of a complex function is equivalent to the limit of two real valued functions and therefore all of the properties that we have for real valued functions are inherited by complex functions. These are covered in the following theorem stated below.

Theorem 3.4.1. Let $X \subseteq \mathbb{C}, f, g : X \to \mathbb{C}$ and let $z_0 \in \mathbb{C}, a, b \in \mathbb{C}$ such that:

$$\lim_{z\to z_0} f(z) = a, \lim_{z\to z_0} g(z) = b,$$

then, for any $k \in \mathbb{C}$, we have that:

1.
$$\lim_{z\to z_0} k = k$$

2.
$$\lim_{z\to z_0} (f(z) + g(z)) = \lim_{z\to z_0} (f(z)) + \lim_{z\to z_0} (g(z)) = a + b$$

3.
$$\lim_{z\to z_0} (f(z) \cdot g(z)) = \lim_{z\to z_0} (f(z)) \cdot \lim_{z\to z_0} (g(z)) = a \cdot b$$

4. for $b \neq 0$,
$$\lim_{z\to z_0} \left(\frac{f(z)}{g(z)}\right) = \frac{\lim\limits_{z\to z_0} (f(z))}{\lim\limits_{z\to z_0} (g(z))} = \frac{a}{b}$$

5.
$$\lim_{z\to z_0} \left(f(z)^k\right] = \lim_{z\to z_0} (f(z))^k = a^k$$

6.
$$\lim_{z\to z_0} \left(k^{f(x)}\right) = k^{\lim\limits_{z\to z_0} (f(z))} = k^a$$

7.
$$\lim_{z\to z_0} (\log_k(f(z))) = \log_k \left(\lim_{z\to z_0} (f(z))\right) = \log_k(a).$$

Proof. The proof of the theorem is similar to that of real variables, and is mentioned in [23, Page 115]. □

Please note that some of these properties are examples of a more general rule which is of the following:

$$\lim_{z\to z_0} (g(f(z))) = g\left(\lim_{z\to z_0} (f(z))\right).$$

Whilst not all functions obey this rule, many of the functions that we are familiar with do. The next section will look at these functions from some special domain, and the underlying property that relates them.

4 Continuity of Functions

In most cases, we are interested about functions $f : U \to \mathbb{C}$, where $U \subseteq \mathbb{C}$. In real analysis , we have been comfortable with subsets such as the open interval $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ and the closed interval $[a, b] = \{x \in \mathbb{R} : a \leqslant x \leqslant b\}$. It turns out that the name "open" and "closed", given to those type of sets, have an analogous definition in $\mathbb{C}$ too, but, as shown in Theorem 1.2.3, we cannot simply order the complex numbers. So we must proceed with caution. Luckily, the ε-neighbourhood, defined in Section 3, provides a good intuition into such constructions.

We will show that, similarly to that of the real numbers, we can still use Weierstrass' Theorem 4.2.12, and the Extreme Value Theorem 4.2.10 when it comes to complex functions.

The study of sets and the continuity of functions defined on those sets is called **topology**. Interested readers can refer to [8, 15] for details.

4.1 Topology of $\mathbb{C}$

In this subsection, we will introduce notions of bounded, unbounded, closed and open sets. These will then be used to introduce the concept of compact, connected sets which will be useful for Theorem (4.2.10), the Extreme Value Theorem for complex functions.

Definition 4.1.1. Let $U \subseteq \mathbb{C}$ be a set. U is called an **open set** if and only if for all $z_0 \in U$, there exists an $\varepsilon > 0$ such that all of the elements in the ε-neighbourhood around z_0 is in U, which means that $D(z_0 : \varepsilon) \subseteq U$. That is, for all $z \in \mathbb{C}$, there exists an $\varepsilon > 0$ such that:

$$\text{If} \quad z \in D(z_0 : \varepsilon) \quad \text{then} \quad z \in U.$$

.

By definition, any ε-neighbourhood around any $z \in \mathbb{C}$ is an open set. In fact, intuitively, we can think of an open set as a surface with its boundary removed, as shown in Figure 6.

c

Figure 6: A geometrical intuition of an open set U. Since, for all of the points in the set, it doesn't have a "boundary", therefore we can find a circle surrounding it.

Example. An open ball, which is defined as $B(0, 1) = \{z \in \mathbb{C} : |z| < 1\}$ is, intuitively, open, since for all $z_0 \in B(0, 1)$, we can find a $D(\varepsilon : z_0)$ such that $D(\varepsilon : z_0) \subset B(0, 1)$. In fact, the ε we chose is $\varepsilon < 1 - |z_0|$.

Definition 4.1.2. Let $U \subseteq \mathbb{C}$ be a set. U is called a **closed set** if and only if its complement U^c is open.

Taking the above example in Figure 7, it is easy to see that a closed set must have a boundary, because its complement then will not have one. Moreover, the "boundary" of a closed set U can be formalized in the following definition:

Definition 4.1.3. Consider any $U \subseteq \mathbb{C}$. A **closure point** of U is a point $z \in \mathbb{C}$ such that for every $\varepsilon > 0$, there exists $u \in U$ such that $u \in D(z : \varepsilon)$, the ε-neighbourhood around z.

As in Figure 7, the "boundary points" are closure points of U, we can further define a notion where we consider all the closure points:

Definition 4.1.4. A **closure** of a set $U \subseteq \mathbb{C}$, denoted by $\overline{U}$, is the set

$$\overline{U} = U \cup \{\ u : u \text{ is a closure point of } U\ \}$$

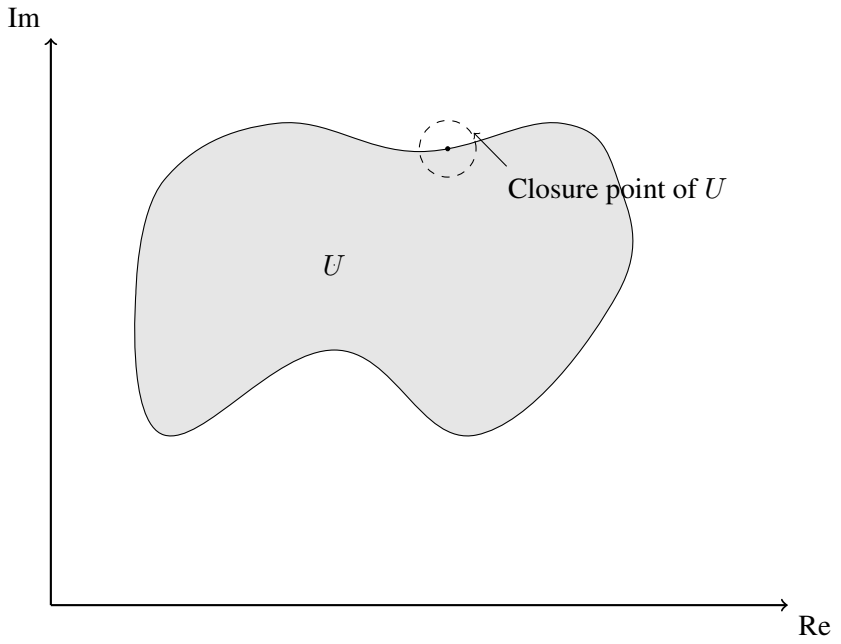


Figure 7: The figure shows a closed set U. If a point z_0 is chosen to be a boundary point, no matter how small ε is chosen, then $D(z_0 : \varepsilon) \not\subset U$. But since the boundary point is contained in an ε-neighbourhood around the point for any ε we choose (we can shrink the circle however we want), it is a closure point of U.

Example. A closed unit disc is defined as

$$B[0,1] = \{z \in \mathbb{C} : |z| \leqslant 1\}$$

. Clearly, no matter how small ε is, if we take $z_0 \in B[0,1]$ such that $|z_0| = 1$, then there exists a $w \in D(z_0 : \varepsilon) \cap B[0,1]^c$. To achieve this, we shall take $w_0 \in D(z_0 : \varepsilon)$ such that $|w_0| = 1 + \varepsilon$.

As its name suggest, the closure of a set takes the boundary of any set and fill it up, so it is easy to show that:

Theorem 4.1.5. A closure of a set is closed.

Proof. Consider a set U and its closure $\overline{U}$. We need to show that $\overline{U}^c$ is open.
If $\overline{U}^c$ is empty, then it is open and we are done. However, if it is not, we can consider any element $x \in \overline{U}^c$. Since by Definition 4.1.4, x is not a limit point of U and $x \notin U$, hence by negating the definition of limit point, there exists $\varepsilon > 0$ such that for all $u \in U$, $u \notin D(x : \varepsilon)$. Hence, for any $x \in \overline{U}^c$ there exists an ε-neighbourhood around x such that $D(x : \varepsilon) \subset \overline{U}^c$. (Since $D(x : \varepsilon)$ does not contain any points and limits points of U). And hence $\overline{U}^c$ is open by definition, as required. □

By Definition 4.1.2 it is intuitive if a set is closed when it is obvious what the complement of the set is, however that is not always the case as can be seen in Example 4.1. Hence, another definition can be introduced.

Theorem 4.1.6 (The Sequential Definition for a Closed Set)**.** A set $U \subseteq \mathbb{C}$ is closed if and only if for all complex sequences $\{z_n\}_{n\in\mathbb{N}}$ such that all of the terms of the sequence are in U, and that $z_n \to w$ implies $w \in U$.

Proof. ($\Rightarrow$:) Assume that the set $U \subseteq \mathbb{C}$ is closed, and that there exists a sequence such that $\lim_{n\to\infty} z_n = w$.
We will proceed by contradiction. Assume that $w \in U^c$. By definition U^c is open. This implies that there exists $\varepsilon > 0$ such that $D(w : \varepsilon) \subset U^c$. Also, since $\lim_{n\to\infty} z_n = w$, there exists a $N \in \mathbb{N}$ such that for all $m \in \mathbb{N}$ such that:

$$\text{if} \quad m \geq N \quad \text{then} \quad z_m \in D(w : \varepsilon),$$

which implies that $z_m \in U^c$. This is a contradiction to the fact that all the terms in the sequence $\{z_n\}_{n\in\mathbb{N}}$ is in U.

($\Leftarrow$:) Now we assume that for all $n \in \mathbb{N}$, $z_n \in U$, if $\lim_{n\to\infty} z_n = w$ then $w \in U$.
Consider the set $\overline{U}$. by Definition 4.1.4, clearly $U \subset \overline{U}$. So it suffices to show that $\overline{U} \subset U$, then $U = \overline{U}$, and hence is closed by Theorem 4.1.5.
For any $z \in \overline{U}$, we can define

$$z_n \in U \cap D\left(z, \frac{1}{n}\right)$$

z_n exists for every n, since z is a limit point, and we can find a point in U for any $\frac{1}{n}$-neighbourhood around z, as shown in Figure 8. Also, since $z_n \to w, w \in U$, we have Therefore, we have $z \in U$, and indeed $\overline{U} \subset U$ and hence we are done.

□

It should be noted that the definition of a set being a closed set is not the negation of a set being an open set and vice versa. This allows a set to be both open and closed at the same time. The following definition is adapted from [17, Chapter 1].

Definition 4.1.7. Let $U \subseteq \mathbb{C}$. U is called a **clopen set** if U is both open and closed.

Example. Examples of clopen sets includes the set $\emptyset$ and $\mathbb{C}$. $\emptyset$ is open since we do not have a $z_0 \in \emptyset$ and so we do not need to check if $D(z_0 : \varepsilon) \subset \emptyset$, where $\varepsilon > 0$. It is closed since there does not exist a sequence $\{z_n\}_{n\in N}$ such that for all $n \in N$, $z_n \in \emptyset$ is convergent. Therefore, we do not need to check whether it converges to an element inside the set.
The set $\mathbb{C}$ is open because for all $z \in \mathbb{C}$, we can take any $\varepsilon > 0$ and we will still have that $D(z : \varepsilon) \subset \mathbb{C}$. It is closed since for all $\{z_n\}_{n\in\mathbb{N}}$ such that for all $n \in N$, $z_n \in \mathbb{C}$ and $\lim_{n\to\infty} z_n = w$, we have that $w \in \mathbb{C}$.

There are some useful properties involving open and closed sets and hence they are quite useful in proofs.

Proposition 4.1.8. Let $U_i, V_i \subseteq \mathbb{C}$ be sets such that for all $i \in \mathbb{N}$, U_i is open and V_i is closed. Then the intersection $\bigcap_{i=1}^{n} U_i$ and the union $\bigcup_{i=1}^{\infty} U_i$ are open sets and $\bigcup_{i=1}^{n} V_i$ and $\bigcap_{i=1}^{\infty} V_i$ are closed sets, where $n \in \mathbb{N}$.

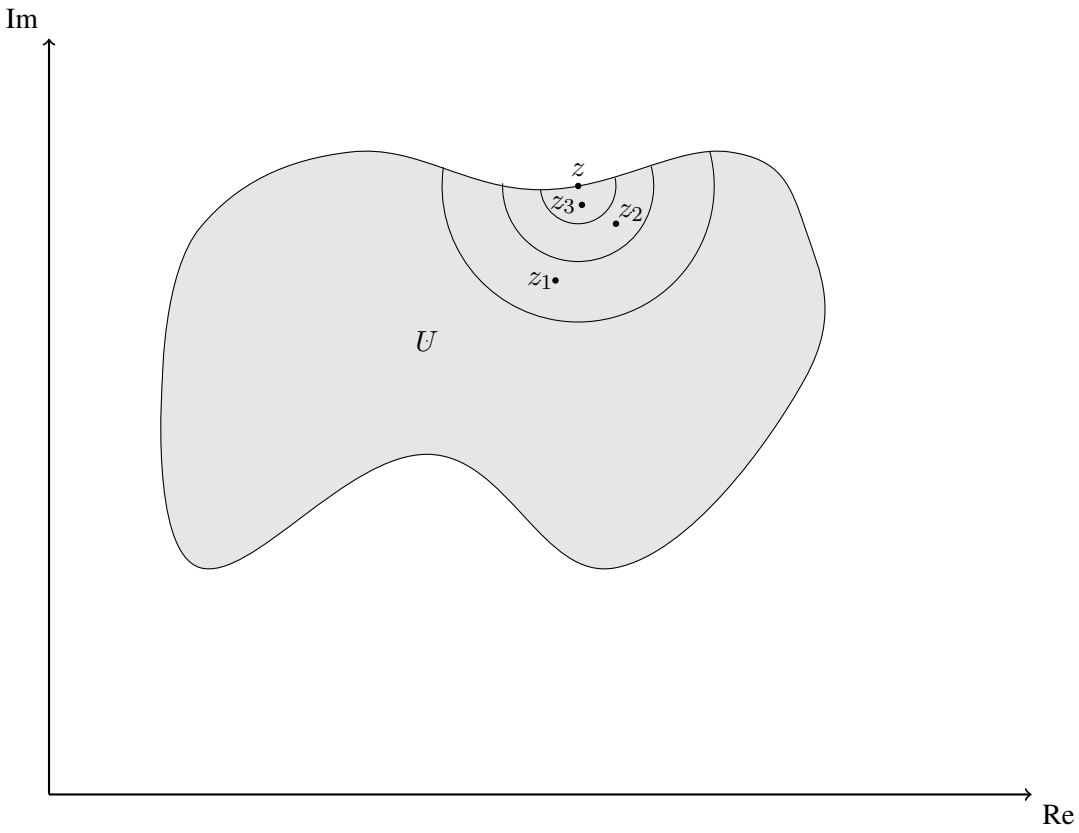


Figure 8: A figure showing how the points $z_1, z_2, z_3...$ are chosen in the proof of Theorem 4.1.6.

Proof. Since, for all positive integers i, U_i is open. Therefore, we have that, for all $z_i \in U_i$, there exists an $\varepsilon_i > 0$ such that all of the elements in the ε_i-neighbourhood around z_i are contained in U_i. Then for all $z_0 \in \bigcup_{i=1}^{\infty} U_i$, there exists a U_i such that $z_0 \in U_i$. This means that there exists ε_i such that it will guarantee that the ε_i-neighbourhood is contained in U_i and hence the ε_i-neighbourhood is in the union $\bigcup_{i=1}^{\infty} U_i$ for a general $z_i \in U_i$. Similarly for $z_0 \in \bigcap_{i=1}^{n} A_i$ there exists an $\varepsilon_0 = \min\{\varepsilon_{i_0}, 0 < i \leqslant n\}$ such that ε_0-neighbourhood is contained within the intersection $\bigcap_{i=1}^{n} U_i$. Hence the intersection $\bigcap_{i=1}^{n} U_i$ and the union $\bigcup_{i=1}^{\infty} U_i$ are open sets.

For $\bigcup_{i=1}^{n} V_i$ and $\bigcap_{i=1}^{\infty} V_i$, consider the complements, which are $\bigcap_{i=1}^{n} V_i^c$ and $\bigcup_{i=1}^{\infty} V_i^c$ respectively by Theorem 1.2.4. Please note that, since for all $i \in \mathbb{N}$, V_i is closed in which implies that V_i^c is open. Hence we see that $\bigcap_{i=1}^{n} V_i^c$ and $\bigcup_{i=1}^{\infty} V_i^c$ are open, and taking the complement again yields that $\bigcup_{i=1}^{n} V_i$ and $\bigcap_{i=1}^{\infty} V_i$ are closed. □

It is sometimes useful to work with subsets rather than sets as can be seen in Theorem 4.1.16 and Corollary 4.2.7, so we shall extend the notion of an open and closed set to subsets.

Definition 4.1.9. Let $U \subseteq \mathbb{C}$ be open. A set $W \subseteq V \subset \mathbb{C}$ is **open relative to** V if $W = V \cap U$.

Definition 4.1.10. Let $U \subseteq \mathbb{C}$ be closed. A set $W \subseteq V \subset \mathbb{C}$ is **closed relative to** V if $W = V \cap U$.

From these definitions, we can deduce the following fact:

Corollary 4.1.11. Let $U, V \subset \mathbb{C}$ such that U is open and V is closed. Then U and V are open and closed relative to themselves, respectively.

Proof. We have that $U = U \cap U$ and $V = V \cap V$. □

Definition 4.1.12. Let $\mathcal{U} = \{U_1, U_2, \ldots\}$, $V \subseteq \mathbb{C}$ be two sets. Then $\mathcal{U}$ is called a **cover** of V if $V \subset \bigcup_{i=1}^{\infty} U_i$. If, for all $i \in \mathbb{N}$, U_i is open, then $\mathcal{U}$ is an **open cover** of V, and if for all $i \in \mathbb{N}$, U_i is closed, then $\mathcal{U}$ is a **closed cover** of V.

Example. Consider the sets $\mathcal{U} = \{U_i : U_i = \{z \in \mathbb{C} : |z| < i\}\}$, $\mathcal{V} = \{V_i : V_i = \{z \in \mathbb{C} : |z| \leqslant i\}\}$ and $W = \{z \in \mathbb{C} : |z| < 10\}$. Then $\mathcal{U}$ is an open cover of W and $\mathcal{V}$ is a closed cover of W.

Definition 4.1.13. Let U be a cover of $W \subseteq \mathbb{C}$. Then $V \subseteq U$ is a **subcover** of W if V is a cover of W. Similarly, if U is an open cover of W, V is an **open subcover** of W, and if U is a closed cover of W, then V is a **closed subcover** of W.

Example. From Example 4.1 above, we know that the sets $\mathcal{U} = \{U_i : U_i = \{z \in \mathbb{C} : |z| < i\}\}$ and $\mathcal{V} = \{V_i : V_i = \{z \in \mathbb{C} : |z| \leqslant i\}\}$ form open and closed covers for $W = \{z \in \mathbb{C} : |z| < 10\}$ respectively. Then U_{11} is an open subcover of W and V_{10} is a closed subcover of W.

Note that subcovers can be either a finite or an infinite collection of sets, but a cover is always infinite. However, in cases when it is obvious, the use of cover is often abused, and hence it is sometimes used to refer to a finite subcover of a given set, since a cover which is infinite almost always exists. Also, a collection of sets that forms a cover of a given set does not mean the same thing as the set being covered by a collection of sets.

Example. The set $\{z : |z| < 1\}$ is covered by the sets $\{z : |z| < 1, \mathrm{Im}(z) > 0\}$ and $\{z : |z| < 1, \mathrm{Im}(z) \leqslant 0\}$, since $\{z : |z| < 1\} = \{z : |z| < 1, \mathrm{Im}(z) > 0\} \cup \{z : |z| < 1, \mathrm{Im}(z) \leqslant 0\}$. However, $\{z : |z| < 1, \mathrm{Im}(z) > 0\}$ and $\{z : |z| < 1, \mathrm{Im}(z) \leqslant 0\}$ do not form a cover of $\{z : |z| < 1\}$.

Definition 4.1.14. Given $W \subseteq \mathbb{C}$, we say that the set W is **disconnected** if and only if there exists two subsets $U, V \subset W$ such that *all* of the following conditions hold:

- U and V are non-empty.
- U and V are open.
- U and V are disjoint (i.e. $U \cap V = \emptyset$).
- U and V covers the set W (i.e. $U \cup V = W$).

Or equivalently, U, V are a non-empty disjoint open cover of W.
The sets U, V are called the **disconnecting sets of** W.
If W is not disconnected, then it is **connected**.

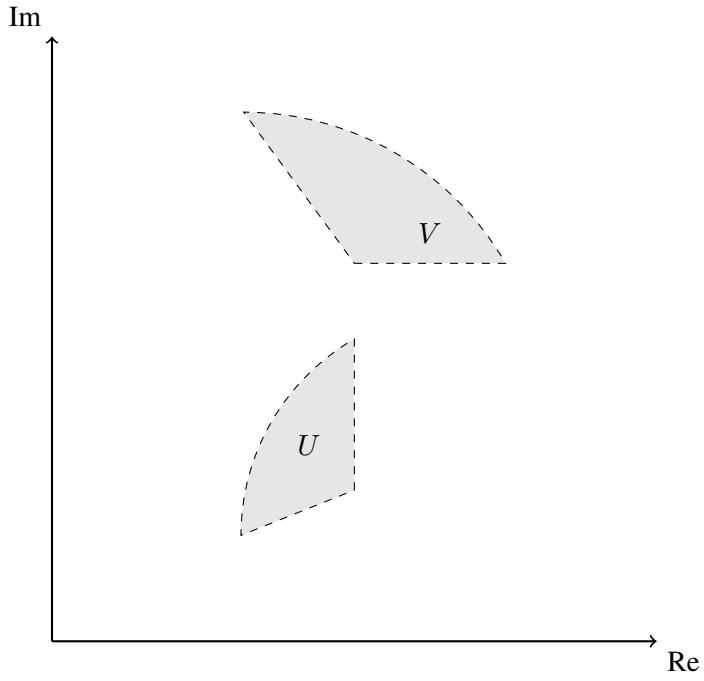


Figure 9: Visualization of a disconnected set, by taking two disjoint open covers U and V.

Proposition 4.1.15. If W is a disconnected set, then W is open.

Proof. By definition there exist two disconnecting sets U and V which form an open cover for W. By Proposition 4.1.8, W is open. □

Theorem 4.1.16. A set U is connected if and only if U and $\emptyset$ are the only sets that are clopen relative to U.

Proof. Firstly, we shall prove that if U is connected, then U and $\emptyset$ are the only sets that are clopen relative to U by contradiction. Assume that $V \subset U$ such that $V \neq \emptyset$ and that V is clopen relative to U. Then, since U is connected and V is clopen relative to U, we have that $U \backslash V$ is also clopen relative to U. Please note that $V \cap U \backslash V = \emptyset$, and hence are disjoint. Also, note that $U = V \cup U \backslash V$ and hence it forms a cover for U. Since V and $U \backslash V$ are clopen relative to U, they are also open relative to U and hence open. Also, since $V \neq \emptyset$, we have that $U \backslash V \neq \emptyset$. So by definition, V and $U \backslash V$ are disconnected sets of U, which is absurd because we had assumed that U is connected. So our assumption that the set $V \subset U$ and $V \neq \emptyset$ must be wrong, leaving the only choice as $V = U$ or $V = \emptyset$.

Conversely, we shall prove the contrapositive. Assume U is disconnected. Then there exists two nonempty, disjoint open covers of U. Let these be V and W. Since V and W are open covers of U, they are open relative to U. Also we have that V and W are closed relative to U, since $V = W^c$ and $W = V^c$, where, by definition, W^c and V^c are closed. Hence V and W are clopen relative to U. Since V and W are non-empty proper subsets of U, we are done. □

Definition 4.1.17. Let $U \subseteq \mathbb{C}$. Then, U is **path-connected** if, for all z, $w \in C$, there exists A function $f : [0, 1] \to \mathbb{C}$ such that f is continuous, $f(0) = z$, $f(1) = w$. If such an f exists, then it is called a **path joining** z **and** w.

Theorem 4.1.18. Let $U \subseteq \mathbb{C}$ be path connected, then U is connected.

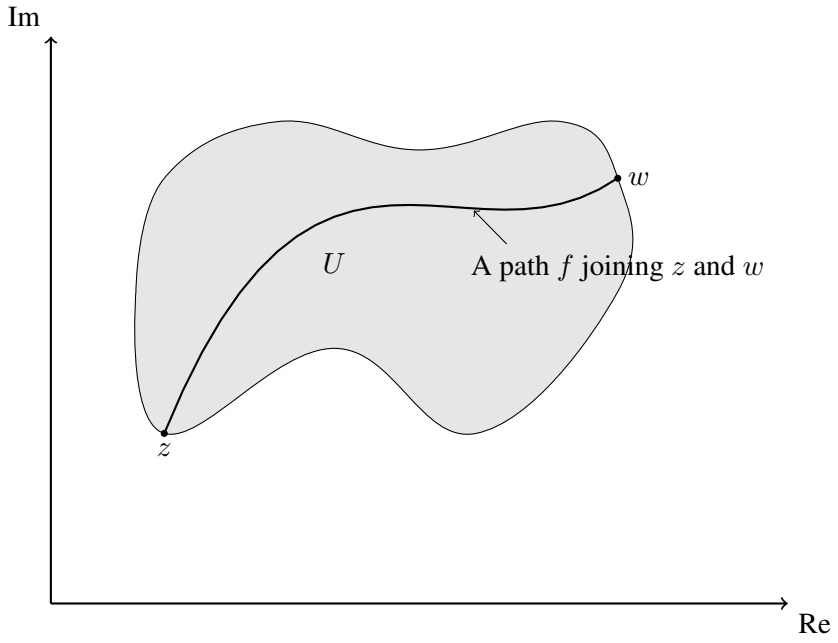


Figure 10: The geometrical intuition of a path-connected set U.

Proof. Let $X \subseteq \mathbb{C}$ be path connected. We proceed to prove by contradiction, assuming that X is disconnected - there exists two proper subsets U and V of X, such that they are disjoint, non-empty, open covers of C. So we can have $V = X \setminus U$. Also, since U, V are non-empty, let $z_1 \in U$ and $z_2 \in V$. Define $f : [0,1] \to X$ to be a continuous path joining z_1 and z_2. Let $W = f^{-1}(U)$. Since f is continuous and by Theorem 4.2.7 and Theorem 4.2.6, U is clopen and W is a clopen subset of $[0,1]$. Since f is a path joining z_1 and z_2, we know that $z_1 \in U$ implies that $0 \in W$ and hence W is non-empty. Similarly, there exists $z_2 \in V$ and so we have that $[0,1] \backslash W \neq \emptyset$. Let $x = \sup W$. Also, since W is clopen, we can deduce that:

- W is closed, hence $x \in W$.
- W is open, hence there exists an ε-neighborhood around x that is contained in W, where $\varepsilon > 0$. This implies that there exists $\varepsilon > 0$ such that $x + \varepsilon \in W$.

Since $x + \varepsilon \in W$, this is a contradiction to the definition of $x = \sup W$. So our assumption that C is disconnected must be wrong, hence C is connected, as required. □

It can also be noted that the converse is not always true.

Example (The Topologist's Sine Curve)**.** Let $U \subset \mathbb{C}$ be defined as the following set:

$$U = \{-1 \leqslant \mathrm{Im}(z) \leqslant 1, 0 < \mathrm{Re}(z) \leqslant 1 : z \in \mathbb{C}\} \cap \left\{ \mathrm{Im}(z) = \sin\left(\frac{1}{\mathrm{Re}(z)}\right) \right\}.$$

Hence, U is connected, but it is not path-connected. This is an important result, as seen in [21, Example 116].

However, the converse will be true if we impose an extra condition:

Theorem 4.1.19. If the set $U \subseteq \mathbb{C}$ is connected and open, then it is path-connected.

Proof. The theorem is a special case mentioned in [7, Theorem 5.12], and we will prove it by contradiction. Let $z_0 \in U$ and assume U is not path-connected. Then there exists

$$V = \{z \in U : z \text{ can be path-connected to } z_0\} \quad W = U \setminus V = \{z \in U : z \text{ cannot be connected to } z_0\} \subseteq U$$

. Please note that $V \neq \emptyset$, since z_0 always can be connected to itself and to $W \neq \emptyset$, since if it was the case, U would be path-connected. Since U is open, for all $w \in V$, there exists an $\varepsilon > 0$ such that $D(w : \varepsilon) \subset U$. Since $w \in V$, there exists a function f path joining w_0 and w, where $w_0 \in D(w : \varepsilon)$. We can therefore choose f to be a straight line. This holds for all $w_0 \in D(w : \varepsilon)$ and for all $w \in V$. This can be repeated until we construct all of U, forcing W to be the empty set. This is because it can be shown by a similar argument that W is open, hence V, W would form a disjoint non-empty open cover of U. But that would lead to a contradiction, since U is connected. Our assumption that U is not path-connected must be wrong, as required.

□

Theorem 4.1.20. Let $f : U \to \mathbb{C}$ be continuous, U is connected and $V \subseteq U$, then $f(V)$ is also connected.

Proof. We shall prove the contrapositive. Assume $f(V)$ is disconnected. Then there exists two disconnecting sets W and X. Since f is continuous, $f^{-1}(W)$ and $f^{-1}(X)$ are disconnecting sets of V by Corollary 4.2.9 and hence, by definition, V is disconnected. □

Definition 4.1.21. A set $U \subseteq \mathbb{C}$ is **bounded** if there exists an $M \in \mathbb{C}$ such that for all $z \in U$, $|z| \leqslant |M|$, and if no such $M \in \mathbb{C}$ exists, then U is **unbounded**.

Example. The set $\{z : |z| < 5\}$ is bounded because by definition, for all $w \in \{z : |z| < 5\}$, $|w| < 5 = |(3, 4)|$.

Another vital concept that is related to sets is the compact set, as it has a very clear connection with the Riemann Sphere, which is introduced in Definition 6.1.3, and hence with the stereographic projection, as well. It is also related with the sequences and uniform continuity.

Definition 4.1.22. A set $U \subset \mathbb{C}$ is compact, if every open cover of U has a finite subcover.

This definition is rather complicated to use as it involves all possible infinite covers. So, in practice, it is much simpler to define compactness using the following theorem:

Theorem 4.1.23 (Heine-Borel Theorem)**.** Let $U \subseteq \mathbb{C}$, and hence the following three conditions are equivalent:

1. U is compact.
2. U is closed and bounded.
3. For complex sequences with all their terms in U, there exists a convergent subsequence which converges to a point in U.

Proof. This is an important theorem, which we will not provide a proof for it. This is mentioned in multiple sources for example in [13, 12]. Note that this is also a generalization of the Bolzano-Weierstrass Theorem in real analysis. □

4.2 Continuity of Complex Functions

In this section, we shall define continuity and the use the compact and connected sets, as defined in the previous section in order to arrive at the Extreme Value Theorem 4.2.10 the Weierstrass' Theorem 4.2.12, and the Riemann Sphere in Section 6.1. This allows us to understand the stereographic projection as an application of complex numbers.

Definition 4.2.1. A function $f : \mathbb{C} \to \mathbb{C}$ is **continuous at** $z_0 \in \mathbb{C}$ if $\lim_{z \to z_0} f(z) = f(z_0)$, where $z \in \mathbb{C}$, or equivalently, for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon.$$

This definition of continuity at a point can be extended to the whole domain of a given complex function.

Definition 4.2.2. A function $f : U \to \mathbb{C}$ is **continuous on** $U \subseteq \mathbb{C}$, if for all $z_0 \in U$, f is continuous at z_0.

It must be noted that the two definitions above are exactly analogous to Definition 3.2.5, in that we can produce the same exact definition by considering the decomposition of the complex function i.e. $f(z) = f(x + iy) = u(x, y) + iv(x, y)$, where x, $y \in \mathbb{R}$, and that u,$v : U$,$V \subseteq \mathbb{R}^2 \to \mathbb{R}$.

Also, considering real analysis, we can define continuity from the perspective of sequences.

Theorem 4.2.3 (Heine's Definition of Continuity)**.** A function $f : U \to \mathbb{C}$ is continuous on $U \subseteq \mathbb{C}$ if and only if for all complex sequences $\{z_n\}_{n \in \mathbb{N}}$ such that all of the terms $z_n \in U$ $\lim_{n \to \infty} z_n = z_0$, $z_0 \in U$ and $\lim_{n \to \infty} f(z_n) = f(z_0)$.

Proof. The proof is similar to that in real analysis and hence trivial. □

It is also possible to define continuity at infinity, but then we have to look whether the function is continuous at $z = \infty$, which is easier to do when considering $z = \frac{1}{w}$, where $w = 0$. Please note that this is closely related to the discussion about limits at infinity as in Theorem 3.3.2.

Definition 4.2.4. A function $f : \mathbb{C} \to \mathbb{C}$ is continuous at $z = \infty$ if $\lim_{z \to 0} f\left(\frac{1}{z}\right) = f\left(\lim_{z \to 0} \frac{1}{z}\right)$.

However, such notions as $z = \infty$ and $z = \frac{1}{w}$, where $w = 0$ are not properly defined in the complex plane and so the extended complex numbers $\mathbb{C} = \mathbb{C} \cup \{\infty\}$ must be introduced, as seen in Subsection 6.1. But, in fact, it is simpler to work with the the Riemann Sphere 6.1.3.

Similar to that of the real-valued functions, continuous functions in the the complex plane have some nice properties. We shall now provide a direct result from Theorem 3.4.1.

Proposition 4.2.5. Let $f : U \to \mathbb{C}$, $g : U \to \mathbb{C}$ and $h : f(U) \to \mathbb{C}$ be continuous functions on U, and let $f(U) \subseteq \mathbb{C}$ respectively, then:

- $f(z) + g(z)$ is a continuous function,
- $f(z)g(z)$ is a continuous function,
- $\frac{f(z)}{g(z)}$ is a continuous function, as long as $g(z) \neq 0, \forall z \in U$,

- $h(f(z))$ is a continuous function.

Proof. Let $f : U \to \mathbb{C}$, $g : U \to \mathbb{C}$ and $h : f(U) \to \mathbb{C}$ be continuous functions on U, and let $f(U) \subseteq \mathbb{C}$ respectively.

Let $\{z_n\}_{n\in\mathbb{N}}$ such that for all $n \in \mathbb{N}$, $z_n \in U$ and that $\lim_{n\to\infty} z_n = z$, where $z \in U$. Since f and g are continuous, $\lim_{n\to\infty} f(z_n) = f(z)$ and $\lim_{n\to\infty} g(z_n) = g(z)$. By Theorem 3.4.1, we have that $\lim_{n\to\infty} f(z_n) + g(z_n) = f(z) + g(z)$ and hence $f(z) + g(z)$ is a continuous function, since this holds for all $z \in U$.

Similarly, by Theorem 3.4.1, $\lim_{n\to\infty} f(z_n)g(z_n) = f(z)g(z)$ and hence $f(z)g(z)$ is a continuous function, since this holds for all $z \in U$.

And again, by Theorem 3.4.1, the statement $\lim_{n\to\infty} \dfrac{f(z_n)}{g(z_n)} = \dfrac{f(z)}{g(z)}$ holds, for all $z \in U$ such that $g(z) \neq 0$ and for all $n \in \mathbb{N}$, $g(z_n) \neq 0$, and hence $\dfrac{f(z)}{g(z)}$ is a continuous function.

Let $z_0 \in U$. Since f is continuous, for all $\varepsilon_1 > 0$, there exists a $\delta_1 > 0$ such that if $|z - z_0| < \delta_1$, then $|f(z) - f(z_0)| < \varepsilon_1$. Since h is continuous on $f(U)$ and $f(z_0) \in f(U)$, therefore for all $\varepsilon > 0$, there exists a $\delta_2 > 0$ such that $|f(z_0) - f(z)| < \delta_2$ implies $|h(f(z_0)) - h(f(z))| < \varepsilon$. Take $\delta = \min\{\delta_1, \delta_2\}$. This holds for all such functions and for all such z_0 in domain of f. □

As stated before, the definition of continuity is equivalent for a complex function that is considered as a sum of two real functions i.e. $f(z) = f(x + iy) = u(x, y) + iv(x, y)$. The sum rule can be used to identify a neat fact that if the complex function is continuous, then the two real functions it is composed of are also continuous. Although this will not be proved, using this fact allows us to show that all fractional polynomials are continuous.

Sometimes it is more convenient to check whether a complex function is continuous indirectly. This can be done using sets.

Theorem 4.2.6. Let $f : \mathbb{C} \to \mathbb{C}$, and let $U, V \subseteq \mathbb{C}$ such that U is closed and V is open. Then, the following statements are equivalent:

1. f is continuous.
2. $f^{-1}(U)$ is closed.
3. $f^{-1}(V)$ is open.

Proof. First we show (1) implies (2). Assume f is continuous. Consider a sequence $\{z_n\}_{n\in\mathbb{N}}$ such that for all $n \in \mathbb{N}$, $z_n \in f^{-1}(U)$ and that $\lim_{n\to\infty} z_n = z$. Since f is continuous, we have that $\lim_{n\to\infty} f(z_n) = f(z)$. Note that for all $n \in \mathbb{N}$,

$$f(z_n) \in U \implies f(z) \in U \implies z \in f^{-1}(U).$$

Therefore, $z \in f^{-1}(U)$ and since f is continuous, we have that $f^{-1}(U)$ is a closed set by Theorem 4.1.6.

Similarly for open sets we have that (1) implies (3).

Now we show that (2) implies (1). Let $z_0 \in f^{-1}(U)$, and let $\varepsilon > 0$ such that $z_0 \in f^{-1}(D(f(z_0), \varepsilon))$ - the ε-neighborhood around $f(z_0)$. Then, there exists a $\delta > 0$ such that the set $D(z_0, \delta)$ is contained in $f^{-1}(D(f(z_0), \varepsilon))$. This is equivalent to saying that there exists a $\delta > 0$ such that $|z - z_0| < \delta$ implies $|f(z) - f(z_0)| < \varepsilon$. And so, by definition, f is continuous.

The same idea works for V^c and so if $f^{-1}(V)$ is open, then f is continuous and $f^{-1}(U)$ is closed implies f is continuous. So (3) implies (1) and we are done. □

In fact, this idea can be extended to complex functions with a more restrictive domain.

Corollary 4.2.7. Let $f : U \to \mathbb{C}$, and let U, V, $W \subseteq \mathbb{C}$ such that V is closed and W is open. Then the following three statements are equivalent:

1. f is continuous,
2. $f^{-1}(V)$ is closed relative to U,
3. $f^{-1}(W)$ is open relative to U.

Proof. The statement is given in [14, Prop. 1.4.10], but the idea of the proof is similar to that of Theorem 4.2.6. □

Theorem 4.2.8. Let the function $f : U \to \mathbb{C}$ be continuous, where U is compact. Then $f(U)$ is also compact.

Proof. Let $f : U \to \mathbb{C}$ be continuous, and let U be compact. Then there exists a open cover $\mathcal{V} = \bigcup_{i=1}^{\infty} V_i$ of U such that $U \subset \bigcup_{i=1}^{n} V_i$, $n \in \mathbb{N}$ and that $\bigcup_{i=1}^{n} V_i$ is a finite open subcover of U. Since f is continuous, $f(U) \subset \bigcup_{i=1}^{n} f(V_i) \subset f(\mathcal{V})$. Hence $f(\mathcal{V})$ is an open cover of $f(U)$ with the set $\bigcup_{i=1}^{n} f(V_i)$ forming a finite open subcover of $f(U)$ which, by definition, means that $f(U)$ is compact. □

Theorem 4.2.9. Let $W \subseteq \mathbb{C}$ be disconnected by the disconnecting sets U and V, and $f : W \to \mathbb{C}$ be continuous. Then $f(W)$ is disconnected by the disconnecting sets $f(U)$ and $f(V)$.

Proof. Since W is disconnected, and by Corollary 4.1.15, we have that W is open. Please note that we can choose the disconnecting sets of W to be U and $V = W \setminus U$. Also, since f is continuous, we have that $f(W) = f(U) \cup f(V)$ with $f(U) \cap f(V) = \emptyset$, where $f(W)$, $f(U)$ and $f(V)$ are open sets, by Theorem 4.2.6. Thus, by definition, $f(W)$ is disconnected by the sets $f(U)$ and $f(V)$. □

Theorem 4.2.10 (Extreme Value Theorem)**.** Let $f : U \to \mathbb{R}$ be continuous, and let $U \subseteq \mathbb{C}$ be compact. Then, there exists z and $w \in U$ such that the function $f(z)$ is the maximum value of the function $f(U)$ and that the function $f(w)$ is the minimum value of the function $f(U)$.

Proof. By Theorem 4.2.8, since f is continuous and U is compact, $f(U)$ is also compact. Using the fact that $f(U)$ is compact, and by Theorem 4.1.23, $f(U)$ is also bounded and closed. Since $f(U)$ is bounded, there exists an $\inf(f(U))$ and a $\sup(f(U)) \in \mathbb{R}$. Since $f(U)$ is closed, we have that $\inf(f(U)), \sup(f(U)) \in f(U)$. This implies that there exists z and $w \in U$ such that $f(z) = \inf(f(U)) = \min(f(U))$ and $f(w) = \sup(f(U)) = \max(f(U))$. □

As can be seen, the notion of a continuous function is extremely useful due to its preservation of certain properties of sets as mentioned above. However, there is another, stronger concept of continuity known as uniform continuity.

Definition 4.2.11. Let $f : U \to \mathbb{C}$, where $U \subseteq \mathbb{C}$. Then f is **uniformly continuous** on U if, for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for $z, w \in U$, if $|z - w| < \delta$, then $|f(a) - f(b)| < \varepsilon$.

This is a stronger notion of continuity as it is a property of a function over the whole domain, rather than a single element of the domain. Also, unsurprisingly, from the definition, it should be clear that every uniformly continuous function is also a continuous function.

Since uniform continuity is easier to work with, as it avoid problems of discontinuity on specific points which can occur for some of the more complicated functions, it can be useful to have an easy way to determine whether a continuous function is uniformly continuous. And in special cases this is quite easy to do.

Theorem 4.2.12 (Weierstrass Theorem)**.** Let $f : U \to \mathbb{C}$ be continuous, and let $U \subseteq \mathbb{C}$ be compact. Then f is uniformly continuous.

Proof. Let $\varepsilon > 0$. Since f is continuous, then for all $z \in U$, there exists a $\delta(z) > 0, w \in U$ such that:

$$|w - z| < \delta(z) \implies |f(w) - f(z)| < \frac{\varepsilon}{2}.$$

Let $\mathcal{D} = \bigcup_{i=1}^{\infty} D_i\left(z_i, \frac{\delta(z_i)}{2}\right)$ be an open cover for U. Indeed this forms an open cover for U, since for all $z \in U$ the neighbourhood $D\left(z, \frac{\delta(z)}{2}\right)$, it is is well-defined and is, by definition, open. Since U is compact, there exists finite $z_1, \ldots, z_n$ such that there exists a finite subcover $\mathcal{D}_n$ of U with the fact that:

$$\mathcal{D}_n = \bigcup_{i=1}^{n} D_k,$$

where $D_k = D\left(z_k, \frac{\delta(z_k)}{2}\right)$, where $1 \leqslant k \leqslant n$. Let $\delta_k = \frac{\delta(z_k)}{2}$. Take $\delta = \min\{\delta_1, \ldots, \delta_n\}$. Then, $|w - z| < \delta$ implies that there exists $1 \leqslant k \leqslant n$ such that $z \in D_k$ and that

$$|z - z_k| < \delta_k \implies |f(z) - f(z_k)| < \frac{\varepsilon}{2}$$

. Also we have that:

$$\begin{aligned} |w - z_k| &= |w - z + z - z_k| \\ &\leqslant |w - z| + |z - z_k| \quad \text{by triangle inequality,} \\ &\leqslant \delta + \delta_k < \delta(z_k) \implies |f(w) - f(z_k)| \leqslant \frac{\varepsilon}{2}. \end{aligned}$$

Hence,

$$\begin{aligned} |f(w) - f(z)| &= |f(w) - f(z_k) + f(z_k) - f(z)| \\ &\leqslant |f(w) - f(z_k)| + |f(z_k) - f(z)| \quad \text{by triangle inequality,} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus, by definition, f is uniformly continuous. □

5 Differentiation in the Complex Plane

Differentiation is an important concept which is used for many applications. For example, it is used to find the gradient for all points of a function, provided it is differentiable in that particular region. We know the concepts of differentiation in the real plane, but we ask ourselves as to what happens if we extend the region into the complex plane. We shall cover important rules and topics that underpin the idea of complex differentiation, and we shall define as to what is a differentiable function in the complex plane, and we shall explore the conditions that a function requires, in order for it to be differentiable in the complex plane.

Firstly, we shall define as to what is differentiation. Some ideas and figures are adapted from [4, Section 2].

5.1 What is Differentiation?

Please note that, for this section, we shall assume that the set U is an open subset of $\mathbb{C}$.

Differentiation is a process in which the ultimate aim is to find a gradient of a particular function at a general point. To understand how to achieve this we must first introduce some useful notation.

Definition 5.1.1. For two points z and $z_0 \in \mathbb{C}$ such that $z_0 \in D(\varepsilon : z)$, where $\varepsilon > 0$, the **change in** z, denoted as Δz, is defined as of the following:

$$\Delta z = z - z_1.$$

Definition 5.1.2. For any function $f : U \to \mathbb{C}$ in the form $w = f(z)$, the **differential coefficient** or the **differential** or more commonly known as the **derivative** of f at point z_0, is given by

$$\frac{dw}{dz} = \frac{df}{dz} = \frac{d}{dz}(f(z_0)) = f'(z_0) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{z \to z_0} \frac{\Delta w}{\Delta z}.$$

.

To understand the above definition, consider two points on the complex plane, one being $(z, f(z))$, and the other being $(z_0, f(z_0))$ within an ε-neighbourhood around z, with $\Delta w = f(z) - f(z_0)$ and $\Delta z = z - z_0$. Now divide Δw with Δz. As z is getting closer and closer, i.e. *limiting*, to z_1, then Δz limits towards 0. This gives $f'(z)$.

Figure 11 describes this process.

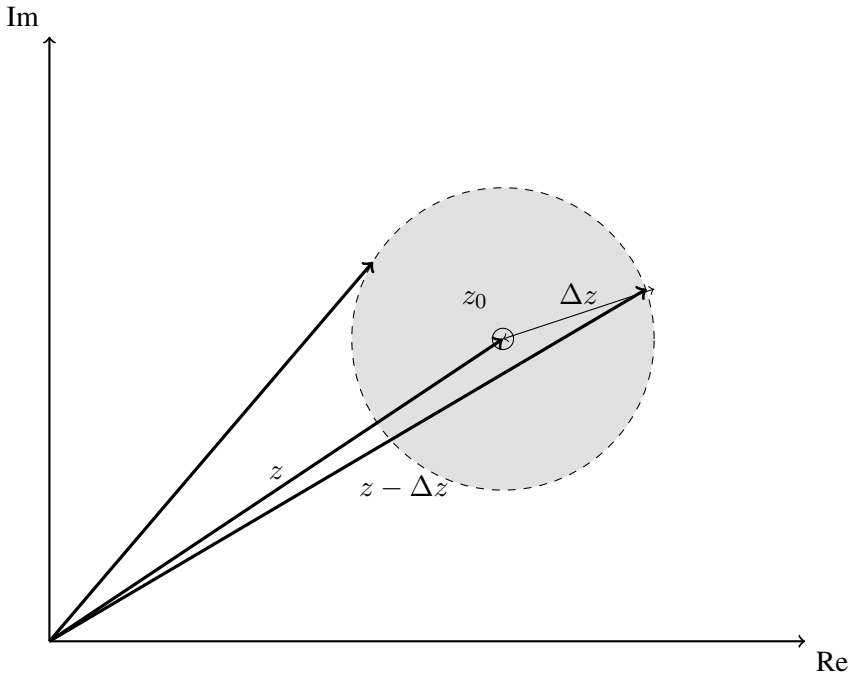


Figure 11: A figure demonstrating differentiation in the complex plane around a point z_1.

Using Definition 5.1.2, we can also deduce that, for a function $f : U \to \mathbb{C}$, the differential coefficient of f at the point z_0 is of the following form:

$$f'(z_0) = \frac{f(z) - f(z_0)}{z - z_0},$$

therefore, we have a new way of defining the derivative:

Definition 5.1.3. The **derivative** of the function $f : U \to \mathbb{C}$ at the point z_0 is defined as the limit, which is stated below:

$$f'(z_0) = \lim_{z \to z_1} \frac{f(z) - f(z_0)}{z - z_0}.$$

From the Definition 5.1.1, we can deduce that $z = z_0 + \Delta z$, and hence we can state yet another definition of the derivative:

Definition 5.1.4. The **derivative** of the function $f : U \to \mathbb{C}$ at the point z_0 is also defined as the following:

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}.$$

Now we can swap out the fixed point z_0 for a general point z, and hence we have a formal definition for the differential coefficient of a complex function at a general point.

Definition 5.1.5. The **derivative** of the function $f : U \to \mathbb{C}$ at a general point z is defined as:

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}.$$

The method of differentiating a function using Definition (5.1.5) is known as "differentiating from first principles".

Now, we shall provide some examples where we will differentiate functions from first principles.

Example. Consider the function $f : \mathbb{C} \to \mathbb{C}$ where $f(z) = z^2 + 5z + 1$. Now, using first principles, we can deduce that:

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{(z + \Delta z)^2 + 5(z + \Delta z) + 1 - z^2 + 5z + 1}{\Delta z} \end{aligned}$$

Using the Binomial Theorem, and expanding out the brackets,

$$f'(z) = \lim_{\Delta z \to 0} \frac{z^2 + 2z\Delta z + (\Delta z)^2 + 5z + 5\Delta z + 1 - z^2 + 5z + 1}{\Delta z}$$

We can cancel out the terms in which the cancellation is shown below.

$$f'(z) = \lim_{\Delta z \to 0} \frac{\cancel{z^2} + 2z\Delta z + (\Delta z)^2 + \cancel{5z} + 5\Delta z + \cancel{1} - \cancel{z^2} + \cancel{5z} + \cancel{1}}{\Delta z}$$

Which leaves us with the equation

$$f'(z) = \lim_{\Delta z \to 0} \frac{2z\Delta z + (\Delta z)^2 + 5\Delta z}{\Delta z}$$

Now we can factorise out the Δz and cancel out the denominator. This is shown below.

$$f'(z) = \lim_{\Delta z \to 0} \frac{\cancel{\Delta z}(2z + \Delta z + 5)}{\cancel{\Delta z}}$$

and hence,

$$f'(z) = \lim_{\Delta z \to 0} 2z + \Delta z + 5.$$

Using Theorem 3.4.1 and Theorem 3.1.1, we can deduce that $f'(z)$ converges to $2z + 5$ as Δz limits towards 0, and hence,

$$f'(z) = 2z + 5.$$

Now, we shall how similar the rules of differentiation of complex functions are to differentiation in the real plane. This will help us differentiate a wide variety of complex functions that are 'differentiable'.

5.2 Basic Rules of Differentiation

If we look at Definition (5.1.5), then we can see that it is very similar from the definition of differentiation in the real plane which is of the following:

Definition 5.2.1. The differential coefficient of the function $f : \mathbb{R} \to \mathbb{R}$ at a general point x is defined as the following:

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Hence the basic rules for differentiating complex functions are going to be very similar to the basic rules for differentiating real functions. That is because the derivation of the formulae uses Definition (5.1.5) for complex functions and Definition (5.2.1) for real functions.

Theorem 5.2.2. The rules for evaluating the differential coefficients of functions, in general, are of the following:

- For a function $f : \mathbb{C} \to \mathbb{C}$, and an arbitrary constant $m \in \mathbb{C}$, we have that:

$$(mf)'(z) = mf'(z).$$

- For two functions $f : \mathbb{C} \to \mathbb{C}$ and $g : \mathbb{C} \to \mathbb{C}$, we have that:
  - $(f + g)'(z)' = f'(z) + g'(z)$,
  - $(f - g)'(z) = f'(z) - g'(z)$,
  - $(f(z)g(z))' = f(z)g'(z) + g(z)f'(z)$ (The Product Rule),
  - $\left(\frac{f(z)}{g(z)}\right)' = \frac{g(z)f'(z) - f(z)g'(z)}{g(z)^2}$ (The Quotient Rule).

Now we are going to look at the chain rule.

Theorem 5.2.3. For two functions $f : \mathbb{C} \to \mathbb{C}$ and $g : \mathbb{C} \to \mathbb{C}$, then, by using the laws of multiplication, we have that:

$$\frac{df}{dz} = \frac{df}{dg} \cdot \frac{dg}{dz},$$

and hence, using the rules of composition and the chain rule mentioned in Theorem (5.2.3), we can deduce that $(f \circ g)' = f'(g(z))g'(z)$.

Having understood how to differentiate it will now be easier to understand what the requirements of a complex function are in order for it to be differentiable.

5.3 Differentiability of Functions

Firstly, we need to define as to what is a "differentiable function". From Definition 5.1.2, we can see that a function, i.e $f : U \to \mathbb{C}$, is differentiable at a point $z \in U$ if and only if the following limit stated below exists.

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}.$$

Hence, by using Definition 5.1.5 we can deduce that:

Definition 5.3.1. A function $f : U \to \mathbb{C}$ is differentiable at the point $z \in U$ if and only if the following limit

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

exists for all $z \in U$.

Hence, we have the following,

Theorem 5.3.2. if $z \in \mathbb{C}$, then the function $f : \mathbb{C} \to \mathbb{C}$ is differentiable if the following limit

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}$$

has the same values when Δz is real as to when Δz is imaginary.

Example. Consider a function $g : \mathbb{C} \to \mathbb{C}$, $g(z) = \overline{z}$. Then the differential coefficient of $g(z)$, if it exists, is of the following form:

$$g'(z) = \lim_{\Delta z \to 0} \frac{g(z + \Delta z) - g(z)}{\Delta z},$$

$$= \lim_{\Delta z \to 0} \frac{\overline{z + \Delta z} - \overline{z}}{\Delta z},$$

And, from the rules of complex conjugation,

$$g'(z) = \lim_{\Delta z \to 0} \frac{\overline{z} + \overline{\Delta z} - \overline{z}}{\Delta z},$$

And cancelling the z terms,

$$g'(z) = \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z}.$$

This means that, by the rules of differentiation,

$$(\overline{z})' = \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z}.$$

and hence,

$$\lim_{\Delta z \to 0} \frac{\Delta \overline{z}}{\Delta z} = \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z}. \tag{5.2}$$

If Δz is real, then $\overline{\Delta z} = \Delta z$, but if Δz is imaginary, then $\overline{\Delta z} = -\Delta z$, and hence, if z is real, then

$$\begin{aligned}\lim_{\Delta z \to 0} \frac{\Delta \overline{z}}{\Delta z} &= \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{\Delta z}{\Delta z} \\ &= \lim_{\Delta z \to 0} 1 \\ &= 1.\end{aligned}$$

However, if Δz is imaginary, then

$$\begin{aligned}\lim_{\Delta z \to 0} \frac{\Delta \overline{z}}{\Delta z} &= \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{-\Delta z}{\Delta z} \\ &= \lim_{\Delta z \to 0} -1 \\ &= -1.\end{aligned}$$

Thus, if we compare the real values of Δz and the imaginary values of Δz, we would get two different values for the derivative at those points. Hence the function $g(z) = \overline{z}$ is not "differentiable", unless if Δz is purely real or if Δz is purely imaginary.

Because any complex variable function, i.e f, can be decomposed into its real and imaginary parts, and following from Section 3, we have that the real and the imaginary parts of a complex function can be expressed as multivariate functions, i.e $f(x + iy) = u(x, y) + iv(x, y)$. This will be useful in the next section.

5.4 Cauchy Riemann Equations

We shall use the multivariate properties of complex functions, i.e. $f(x + iy) = u(x, y) + iv(x, y)$ to derive partial derivatives. Firstly, we need to define as to what is $f(z + \Delta z)$.

Definition 5.4.1. Consider a function $f : \mathbb{C} \to \mathbb{C}$ and the two multivariate functions $u : \mathbb{R}^2 \to \mathbb{R}$; $(x, y) \to u(x, y)$ and $(x, y) \to v(x, y)$, and $v : \mathbb{R}^2 \to \mathbb{R}$. Hence, $f(z + \Delta z)$ is defined as of the following:

$$\begin{aligned}f(z + \Delta z) &= f(x + \Delta x + i(y + \Delta y)) \\ &= u(x + \Delta x) + iv(y + \Delta y).\end{aligned}$$

From Definition 5.1.5 we can deduce that:

Theorem 5.4.2. For a function $f : \mathbb{C} \to \mathbb{C}$ and the two multivariate functions $u : \mathbb{R}^2 \to \mathbb{R}$, and $v : \mathbb{R}^2 \to \mathbb{R}$;

$(x, y) \to u(x, y)$ and $(x, y) \to v(x, y)$, then the derivative of f can be expressed as of the following:

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y} \\ &= \lim_{\Delta u, \Delta v \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \end{aligned}$$

If Δz is real, then $\Delta y = 0$, and hence $f'(z)$ is of the following form:

$$\begin{aligned} f'(z) &= \lim_{\Delta u, \Delta v \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \\ &= \lim_{\Delta u, \Delta v \to 0} \frac{u(x + \Delta x, y) + iv(x + \Delta x, y) - u(x, y) - iv(x, y)}{\Delta x} \\ &= \lim_{\Delta u \to 0} \left(\frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \right) \end{aligned}$$

And, by definition, $f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x}$.

Using a similar method above, but Δz is imaginary ($\Delta x = 0$), we get that $f'(z) = \dfrac{1}{i}\dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial y} = \dfrac{\partial v}{\partial y} - i\dfrac{\partial u}{\partial y}$.

In order for it to be differentiable, $f'(z)$ has to be the same, no matter what Δz is, as mentioned in Theorem 5.3.2. Therefore we have that:

$$\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

and hence, by equating the real parts and the imaginary parts, we have a following theorem:

Theorem 5.4.3 (Cauchy Riemann Equations)**.** Consider a function $f : \mathbb{C} \to \mathbb{C}$ and the two multivariate functions $u : \mathbb{R}^2 \to \mathbb{R}$; $(x, y) \mapsto u(x, y)$ and $v : \mathbb{R}^2 \to \mathbb{R}$; $(x, y) \mapsto v(x, y)$. If the function f is differentiable then:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

6 Application: Geometry of Complex Numbers

6.1 The Stereographic Projection

As mentioned in Section 1, the stereographic projection is a function from the sphere to a plane. In the context of complex analysis, it allows us to introduce a very important concept called the extended complex plane, in which is defined below.

Definition 6.1.1. The **extended complex plane** is defined as the set $\overline{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.

It is very useful for the study of limits, series and functions in general as it allows the concept of infinity to be treated as an another well defined point.

However, the introduction of a point called the point at infinity does not get rid of all of the indeterminate forms.

Definition 6.1.2 (Operation in $\overline{\mathbb{C}}$)**.** $\overline{\mathbb{C}}$ takes the same rules of arithmetic as $\mathbb{C}$, except at the point ∞. Let z, $w \in \mathbb{C}$ such that $w \neq 0$, then

$$\begin{aligned} z + \infty &= \infty \\ \infty + \infty &= \infty \\ w \cdot \infty &= \infty \\ \infty \cdot \infty &= \infty \\ \frac{z}{\infty} &= 0 \end{aligned}$$

but $\infty - \infty$, $\frac{\infty}{\infty}, 0 \cdot \infty$ are undefined, they are often refereed as "indeterminate forms".

The basic idea behind projecting every point from a sphere to a plane can be achieved using a sphere positioned anywhere on the plane with either a single point of intersection between the sphere and the plane, a locus of points (forming a circle on the plane) or no intersection at all (the sphere is above or below the plane). Perhaps the most famous two ways of positioning the sphere is either with a single point intersection at the origin or in a way that forms a Riemann Sphere, as shown in Figure 12.

Definition 6.1.3. A **Riemann Sphere** is a unit sphere centered at origin, denoted by $\mathbb{S}^2$, with the formula

$$\mathbb{S}^2 = \left\{\xi^2 + \eta^2 + \zeta^2 : \xi, \eta, \zeta \in \mathbb{R}\right\}$$

As stated above, the positioning of the sphere and its radius is not important for the mapping, but it does make the mapping easier to understand, visualise and use if an appropriate positioning and size are selected and hence $\mathbb{S}^2$ will be used in examples.

Definition 6.1.4. The **stereographic projection** is a bijective function $S : \overline{\mathbb{C}} \to \mathbb{S}^2$ given by

$$S(w) = \begin{cases} \left(\frac{2\mathrm{Re}(w)}{|w|^2+1}, \frac{2\mathrm{Im}(w)}{|w|^2+1}, \frac{|w|^2-1}{|w|^2+1}\right) & w \in \mathbb{C} \\ (0,0,1) & w = \infty \end{cases}$$

.

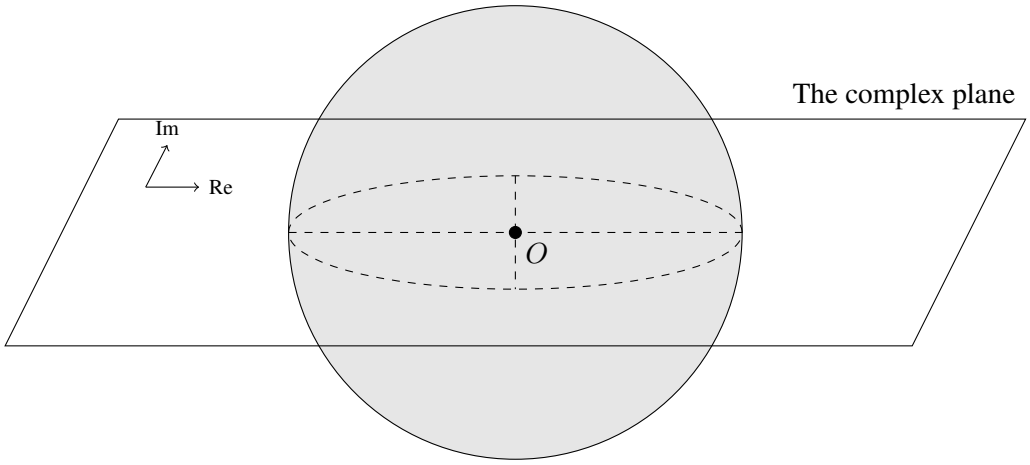


Figure 12: Depiction of the Riemann Sphere $\mathbb{S}^2$, centered at the origin.

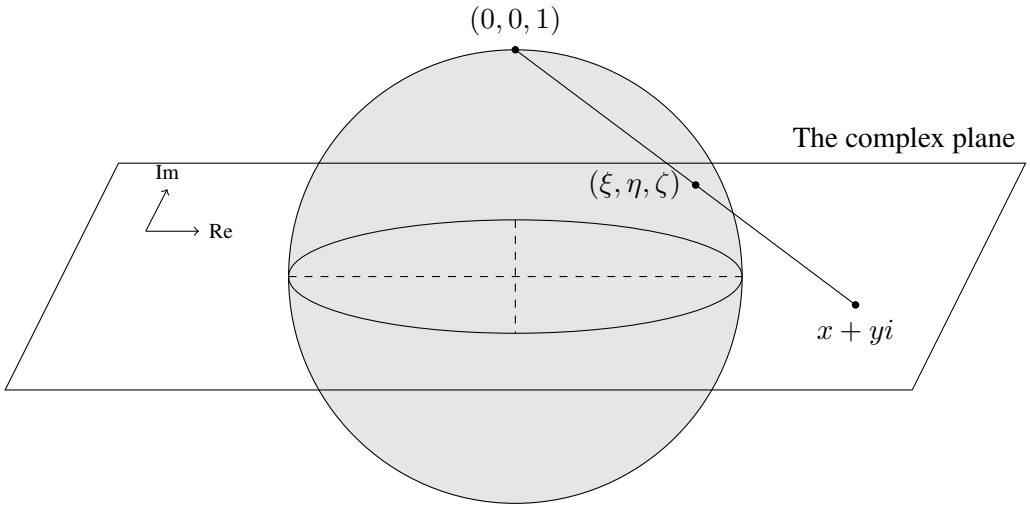


Figure 13: Stereographic projection.

To understand the derivation, the projection is depicted in Figure 13:

Let N be the North pole with coordinates $(0, 0, 1)$, $w \in \mathbb{C}$ with coordinates (x, y) and the point P with coordinates (ξ, η, ζ), with $\xi^2 + \eta^2 + \zeta^2 = 1$. Note that we are working over the field of $\mathbb{R}$ and so $x, y, \xi, \eta, \zeta \in \mathbb{R}$. From the diagram we construct the Cartesian equation of the line in three dimensions, passing through N and w.

$$\frac{\xi}{x} = \frac{\eta}{y} = 1 - \zeta = \lambda \tag{6.3}$$

$\lambda \in \mathbb{R}$. Since $|\mathbb{S}^2| = 1$ we have to take $\lambda = \frac{2}{|w|^2+1}$ or $\lambda = 0$ in order for ξ, η and $\zeta \in \mathbb{S}^2$. Taking $\lambda = 0$ gives $N = (0, 0, 1)$, which we exclude since $w \neq \infty$. And taking $\lambda = \dfrac{2}{|w|^2 + 1}$ gives

$$P = \left(\frac{2x}{|w|^2 + 1}, \frac{2y}{|w|^2 + 1}, \frac{|w|^2 - 1}{|w|^2 + 1} \right)$$

, which is the desired result.

Theorem 6.1.5. Stereographic projection is bijective.

Proof. We shall first prove injectivity. Note that by definition of S if we choose $z, z' \in \overline{\mathbb{C}}$ such that $S(z) = S(z') = \infty$, then it is trivial. So let $w, w' \in \mathbb{C}$ such that $S(w) = S(w')$. Then,

$$\begin{gathered} S(w) = S(w') \\ \left(\frac{2\text{Re}(w)}{|w|^2+1}, \frac{2\text{Im}(w)}{|w|^2+1}, \frac{|w|^2-1}{|w|^2+1}\right) = \left(\frac{2\text{Re}(w')}{|w'|^2+1}, \frac{2\text{Im}(w')}{|w'|^2+1}, \frac{|w'|^2-1}{|w'|^2+1}\right). \end{gathered}$$

Comparing entries we have that,

$$\begin{aligned} \frac{|w|^2-1}{|w|^2+1} &= \frac{|w'|^2-1}{|w'|^2+1} \\ \left(|w|^2-1\right)\left(|w'|^2+1\right) &= \left(|w'|^2-1\right)\left(|w|^2+1\right) \\ |w|^2|w'|^2 + |w|^2 - |w'|^2 - 1 &= |w|^2|w'|^2 + |w'|^2 - |w|^2 - 1 \\ 2|w|^2 &= 2|w'|^2 \\ |w| &= |w'|. \end{aligned}$$

This allows us to conclude that,

$$\begin{aligned} \frac{2\text{Re}(w)}{|w|^2+1} &= \frac{2\text{Re}(w')}{|w'|^2+1} \\ \frac{2\text{Re}(w)}{|w|^2+1} &= \frac{2\text{Re}(w')}{|w|^2+1} \\ \text{Re}(w) &= \text{Re}(w') \end{aligned}$$

and similarly,

$$\begin{aligned} \frac{2\text{Im}(w)}{|w|^2+1} &= \frac{2\text{Im}(w')}{|w'|^2+1} \\ \frac{2\text{Im}(w)}{|w|^2+1} &= \frac{2\text{Im}(w')}{|w|^2+1} \\ \text{Im}(w) &= \text{Im}(w'). \end{aligned}$$

So $w = w'$, and so S is injective.

Now we shall prove surjectivity. Let $(\xi, \eta, \zeta) \in \mathbb{S}^2$ such that $(\xi, \eta, \zeta) \neq (0, 0, 1)$, since that case is trivial. Then there exists $w \in \mathbb{C}$ given by $\left(\frac{\xi}{1-\zeta}, \frac{\eta}{1-\zeta}\right)$ such that $S(w) = (\xi, \eta, \zeta)$. This holds for all such $(\xi, \eta, \zeta) \in \mathbb{S}^2$ since $\zeta = 1$ implies that $(\xi, \eta, \zeta) = (0, 0, 1)$, and hence S is surjective.

□

Since we now know that stereographic projection is in fact a bijection we can define its inverse mapping.

Definition 6.1.6. The **inverse stereographic projection** is $S^{-1} : \mathbb{S}^2 \to \overline{\mathbb{C}}$ given by

$$S^{-1}(\xi, \eta, \zeta) = \left(\frac{\xi}{1-\zeta}\right) + \left(\frac{\eta}{1-\zeta}\right) i$$

.

To derive the mapping from the sphere to the plane we can consider the same set-up of N and $P \in \mathbb{S}^2$ and $w \in \mathbb{C}$. This time the equation of the line connecting N and P is given by

$$\frac{x}{\xi} = \frac{y}{\eta} = \frac{z-1}{\zeta - 1} = \lambda. \tag{6.4}$$

We know that $w = (x, y, 0)$, which gives that $\lambda = \dfrac{1}{1-\zeta}$.

Having understood the definition of S, we can explore its behaviour.

Theorem 6.1.7. Consider the ε-neighbourhood around $z = \infty$ (introduced in definition), the image of the set after stereographic projection is a small circle around the north pole on $\mathbb{S}^2$.

Proof. (Informal.) We haven't introduced how to rigorously define a small circle on the sphere, so it is hard to provide a complete analytical proof. However, one can see a intuitive explanation of the theorem using the Figure 14 below: □

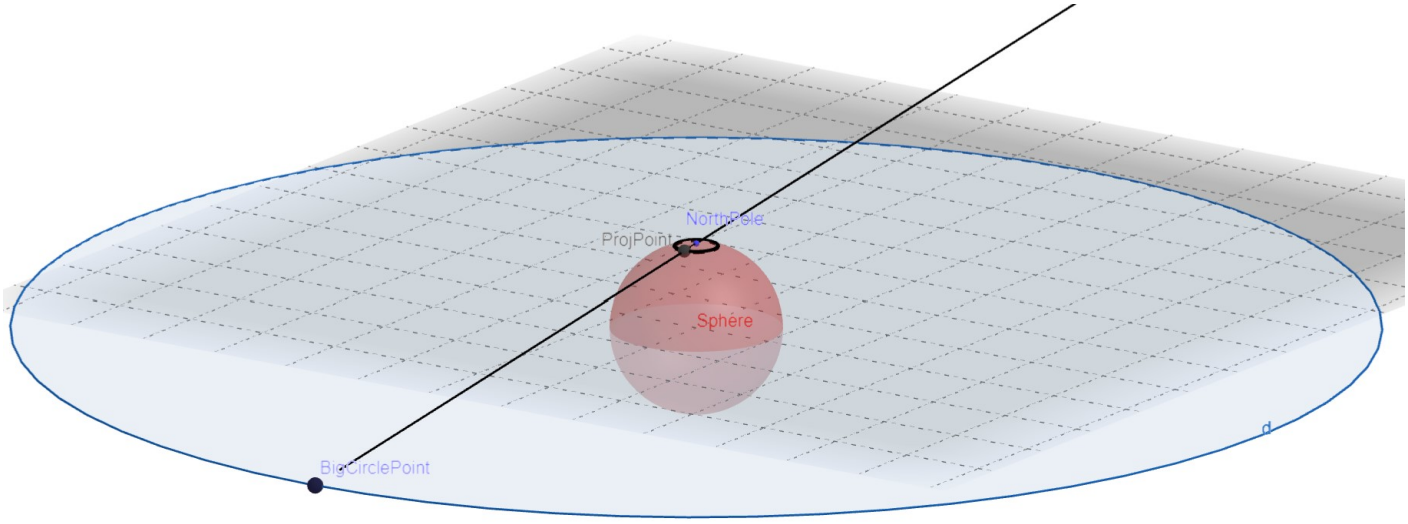


Figure 14: A figure showing the visualization of the neighbourhood around infinity, which is the plane **not** highlighted in blue circle. While $\mathbb{S}^2$ is in red, the stereographic projection denoted by the black line did project the infinity neighbourhood into a small circle around the north pole.

Stereographic projection is very useful in context of complex function mapping, for example as mentioned in Theorem 6.2.1

6.2 Geometry of the extended Complex plane

Given $f : \overline{\mathbb{C}} \to \overline{\mathbb{C}}$, it is often convenient to think of it as a mapping which sends a complex number to another complex number, with the rules of treating the point ∞ defined above in Definition 6.1.2. In the following section, we introduce a few function which have geometric significance.

Example. Translation: $f(z) = \mathcal{T}_b(z) = z + b$ for some complex constant b.
Since complex addition is similar to vector addition in $\mathbb{R}^2$, this function takes every complex number as a $\mathbb{R}^2$ vector and translate them by adding the vector representing the complex number b. This can be shown in Figure 15:

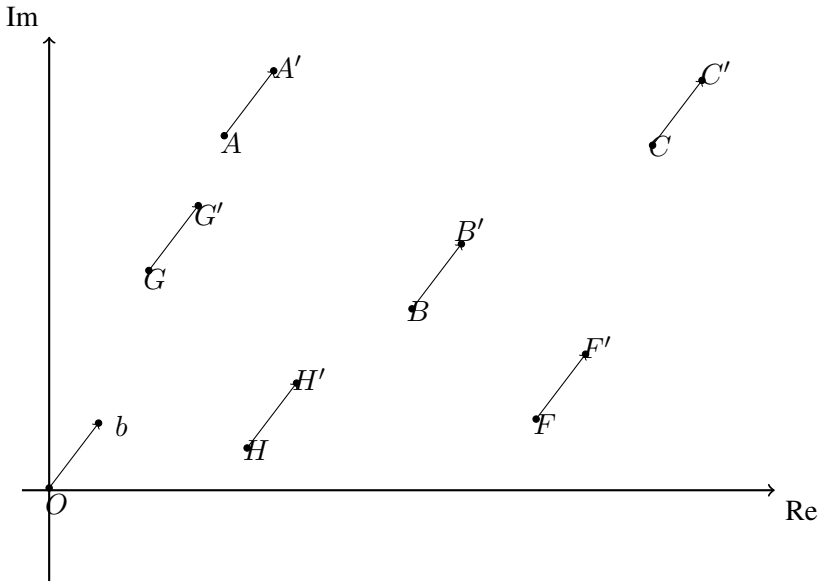


Figure 15: The effect of the translation map $\mathcal{T}_b$ by some complex number b. For example, the point A is mapped to A'.

Example. Multiplication (or Dilation): $f(z) = \mathcal{D}_c(z) = cz$ for some non-zero complex constant c. Note that we can write the two complex number in their polar form: $c = r_1 e^{i\theta_1}, z = re^{i\theta}$, then

$$cz = r_1 e^{i\theta_1} re^{i\theta} = r_1 \left(re^{i(\theta_1 + \theta)} \right)$$

So given a complex number z, $\mathcal{D}$ rotate the vector by $\arg(c) = \theta_1$, then scale the vector by a factor of r_1. A visualization can be seen on Figure 16

Example. Reciprocal: $f(z) = \mathcal{R}(z) = \dfrac{1}{z}$.

It might not be apparent that this map have geometric significance, but the effect of this map is, surprisingly, related to the stereographic projection introduced above at Section 6.1:

Theorem 6.2.1. If for all $w, z \in \mathbb{C}$ we have $w = \dfrac{1}{z}$, then the stereographic projection S introduced in Definition 6.1.4 would map w and z to diametrically opposite points on the Riemann Sphere.

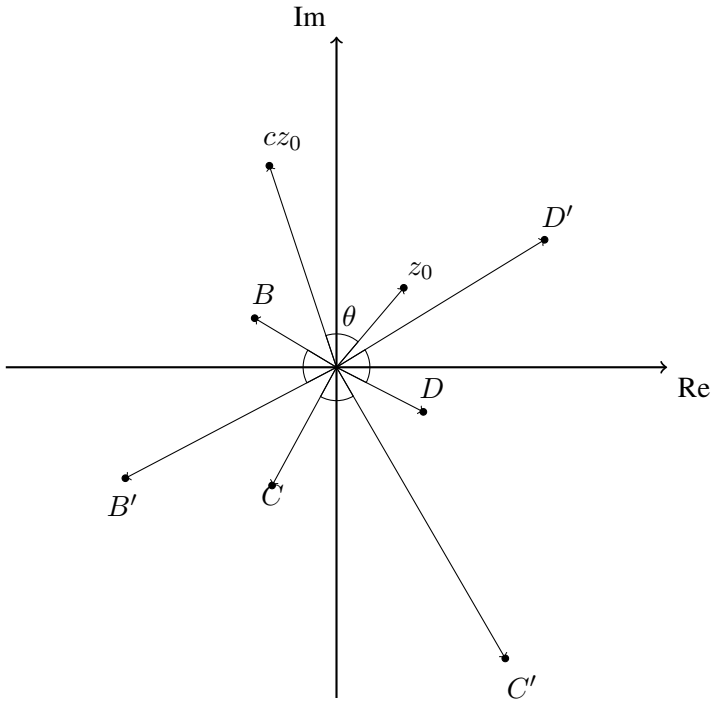


Figure 16: A Visualization of the dilation map $\mathcal{D}_c$ by some complex number c. Note that the complex number z_0 is rotated anti-clockwise by $\arg(c)$ and stretch by a factor of $|c|$.

Proof. This statement is introduced in [3, Lemma 13.9.2], for which we would now provide a proof on.
Since the stereographic projection is bijective, it is equivalent to show that diametrically opposite points map to two complex number with the relation $w = \dfrac{1}{z}$. As seen in the Figure 17, consider any point $P = (\xi, \eta, \zeta) \in \mathbb{S}^2$, then it's diametrically opposite point Q have coordinate $(-\xi, -\eta, -\zeta)$. So we can consider its inverse stereographic projection $S^{-1}(P)$ and $S^{-1}(Q)$, given by Definition 6.1.6. Hence:

$$S^{-1}(P) = \frac{\xi}{1-\zeta} + \frac{\eta}{1-\zeta}i, \quad S^{-1}(Q) = \frac{(-\xi)}{1-(-\zeta)} + \frac{-\eta}{1-(-\zeta)}i = -\frac{\xi}{1+\zeta} - \frac{\eta}{1+\zeta}i$$

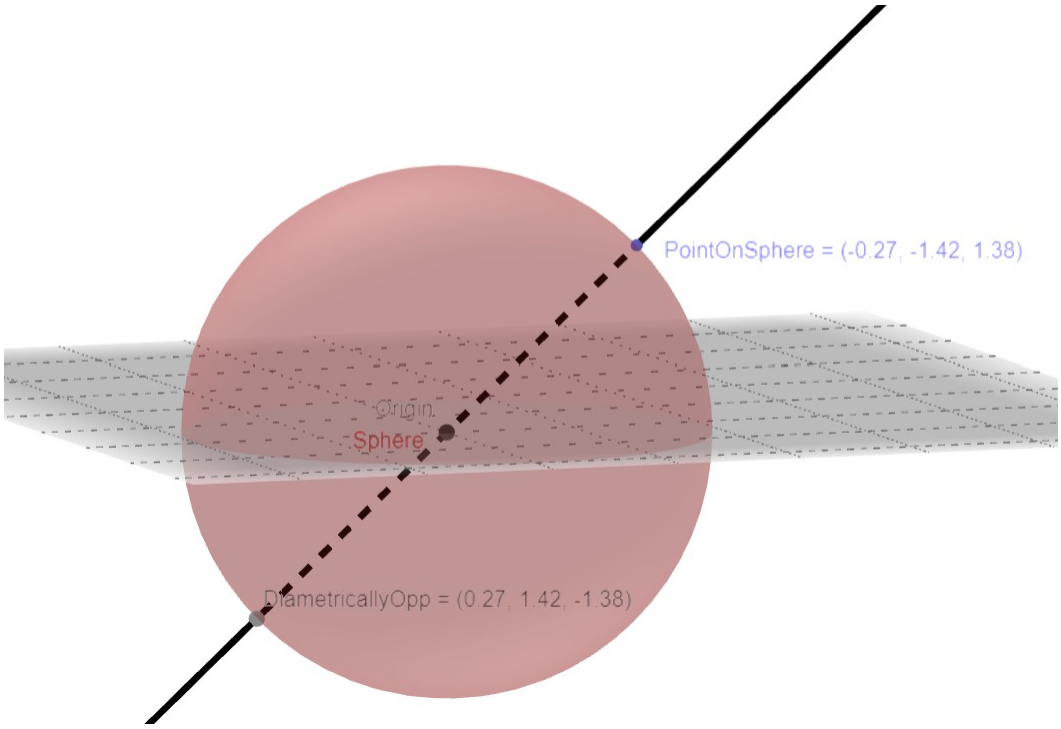


Figure 17: Consider the point `PointOnSphere` having a coordinate of (ξ, η, ζ) on $\mathbb{S}^2$, it is obvious that its diametrically opposite point have an coordinate of $(-\xi, -\eta, -\zeta)$

Hence we have, using the fact that $\xi^2 + \eta^2 = 1 - \zeta^2$, since $(\xi, \eta, \zeta) \in \mathbb{S}^2$:

$$\begin{aligned}
\frac{1}{S^{-1}(P)} &= \frac{1}{\frac{\xi}{1-\zeta} + \frac{\eta}{1-\zeta}i} \\
&= \frac{\left(\frac{\xi}{1-\zeta} - \frac{\eta}{1-\zeta}i\right)}{\left(\frac{\xi}{1-\zeta} + \frac{\eta}{1-\zeta}i\right)\left(\frac{\xi}{1-\zeta} - \frac{\eta}{1-\zeta}i\right)} \\
&= \frac{\left(\frac{\xi}{1-\zeta} - \frac{\eta}{1-\zeta}i\right)}{\left(\frac{\xi}{1-\zeta}\right)^2 + \left(\frac{\eta}{1-\zeta}\right)^2} \\
&= \frac{\left(\frac{\xi}{1-\zeta} - \frac{\eta}{1-\zeta}i\right)}{\frac{\xi^2+\eta^2}{(1-\zeta)^2}} \\
&= \frac{(1-\zeta)(\xi - \eta i)}{\xi^2 + \eta^2} \\
&= \frac{(1-\zeta)(\xi - \eta i)}{1 - \zeta^2} \\
&= \frac{(\xi - \eta i)}{1+\zeta} = S^{-1}(Q)
\end{aligned}$$

As required. □

Hence, a geometrical representation of the mapping $\mathcal{R}(z)$ can be visualized in the Figure 18:

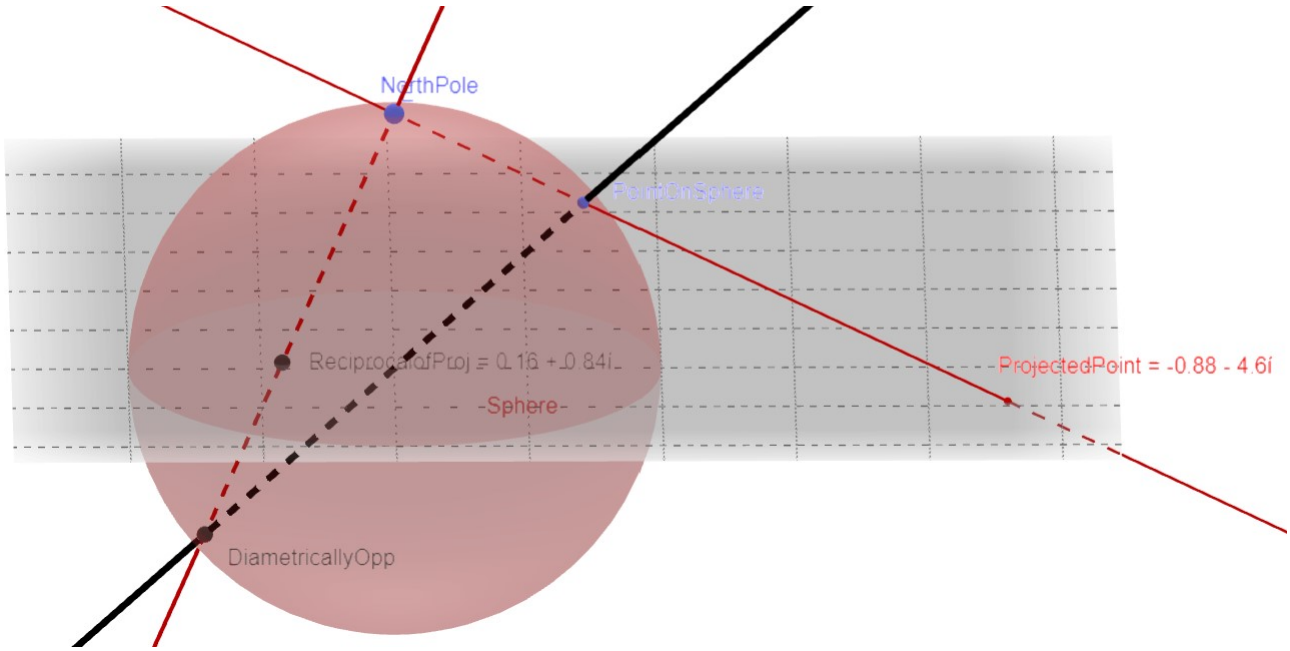


Figure 18: A geometrical visualization of the map $\mathcal{R}(z) = \frac{1}{z}$ via the stereographic projection. If we consider a point on the sphere and its diametrically opposite point, then its inverse stereographic projection S^{-1} will bring them to the reciprocal of each other.

We will now introduce a more general type of map called the Möbius maps:

Definition 6.2.2. A **Möbius Map** is a function $f : \mathbb{C} \to \mathbb{C}$ with the following form:

$$f(z) = \frac{az + b}{cz + d}$$

with $ad - bc \neq 0$.

It is easy to see this notion generalized the above three map:

- Translation is when $c = 0, a = d = 1$.
- Multiplication is when $b = c = 0, d = 1$.
- Inverse is $a = d = 0, b = c = 1$.

Theorem 6.2.3. A circle in complex plane have the form

$$A = \{z \in \mathbb{C} : az\overline{z} + b\overline{z} + \overline{b}z + c = 0\} \tag{6.5}$$

with $a, c \in \mathbb{R}, b \in \mathbb{C}, |b|^2 - ac > 0$, and the center of A is the complex number $\frac{-b}{a}$.

Proof. Note that for all complex number a, we have $|a|^2 = a\overline{a}$,

$$\begin{aligned} az\overline{z} + b\overline{z} + \overline{b}z + c &= 0 \\ a^2 z\overline{z} + ab\overline{z} + a\overline{b}z + ac &= 0 \\ a^2 z\overline{z} + ab\overline{z} + a\overline{b}z + ac + |b|^2 &= |b|^2 \\ a^2 z\overline{z} + ab\overline{z} + a\overline{b}z + |b|^2 &= |b|^2 - ac \\ a^2 z\overline{z} + ab\overline{z} + a\overline{b}z + \overline{b}b &= |b|^2 - ac \\ a\overline{z}(az+b) + \overline{b}(az+b) &= |b|^2 - ac \\ (az+b)(a\overline{z} + \overline{b}) &= |b|^2 - ac \\ |az+b|^2 = |b|^2 - ac \qquad \text{so} \qquad |az+b| &= \sqrt{|b|^2 - ac} \end{aligned}$$

This is hence the circle-radius form we had hopep for, $\left|z + \dfrac{b}{a}\right| = \dfrac{\sqrt{|b|^2 - ac}}{a}$, and hence the proof is completed. □

Theorem 6.2.4. The three maps above $\mathcal{T}, \mathcal{D}$ and $\mathcal{R}$ send a circle to a circle and so does the composition of them.

Proof. It suffices to show the first part of the statement, since composition preserve circles will automatically follows. It geometrically obvious that scaling (that is, dilation) translation sends circles to circles. However, to see the map $f(z) = \dfrac{1}{z}$ sends circle to circle, we have to show that (for all a, b, c) after the substitution $z \to \dfrac{1}{z}$, we can still write down (6.5) as:

$$dz\overline{z} + \overline{e}z + e\overline{z} + f = 0 \qquad \text{for some} \qquad d, f \in \mathbb{R}, e \in \mathbb{C}, |e|^2 - df > 0 \tag{6.6}$$

So, after the substitution we have:

$$\begin{aligned} a\frac{1}{z}\overline{\left(\frac{1}{z}\right)} + \overline{b}\frac{1}{z} + b\overline{\left(\frac{1}{z}\right)} + c &= 0 \\ a\frac{1}{z}\left(\frac{1}{\overline{z}}\right) + \overline{b}\frac{1}{z} + b\left(\frac{1}{\overline{z}}\right) + c &= 0 \\ \frac{a}{z\overline{z}} + \frac{\overline{b}}{z} + \frac{b}{\overline{z}} + c &= 0 \end{aligned}$$

Multiplying the whole equation by $z\overline{z}$ we have that $a + \overline{b}\overline{z} + bz + cz\overline{z} = 0$. Rearranging gives,

$$cz\overline{z} + bz + \overline{b}\overline{z} + a = 0 \tag{6.7}$$

Comparing the above equation with (6.5), we indeed have $d = c, e = \overline{b}, f = a$. Note that the conditions

$$d, f \in \mathbb{R}, e \in \mathbb{C}, |e|^2 - df > 0$$

are automatically satisfied, hence completing the proof. □

Corollary 6.2.5. Möbius mapping f sends generalized circle to generalized circles.

Proof. Note that a, c can't be 0 at the same time due to Möbius mapping requires the condition $ad - bc \neq 0$. Similarly, b, d also can't be simultaneously 0.
If $a = 0$, we must have $b, c \neq 0$ and $f(z) = \dfrac{b}{cz + d} = \mathcal{D}_b \circ \mathcal{R} \circ \mathcal{T}_d \circ \mathcal{D}_c$.

If $c = 0$, we must have $a, d \neq 0$ and $f(z) = \dfrac{az+b}{d} = \mathcal{D}_{\frac{a}{d}} \circ \mathcal{T}_{\frac{b}{d}}$.
In any other cases, observe that

$$\begin{aligned} f(z) &= \frac{az+b}{cz+d} \\ &= \left(\frac{a}{c}\right) \frac{\left(\frac{c}{a}\right)(az+b)}{(cz+d)} \\ &= \left(\frac{a}{c}\right) \frac{cz + \frac{cb}{a}}{cz+d} \\ &= \left(\frac{a}{c}\right) \frac{cz + \frac{cb}{a} + d - d}{cz+d} \\ &= \left(\frac{a}{c}\right) \left(1 + \frac{\frac{cb}{a} - d}{cz+d}\right) \\ &= \mathcal{D}_{\frac{a}{c}} \circ \mathcal{T}_1 \circ \mathcal{D}_{\frac{cb}{a}-d} \circ \mathcal{R} \circ \mathcal{T}_d \circ \mathcal{D}_c(z) \end{aligned}$$

It is easy to check the dilation map $\mathcal{D}$, in all cases, never takes value zero as its argument and is hence well-defined. Since we have shown that all Möbius maps are a composition of $\mathcal{T}, \mathcal{D}$ and $\mathcal{R}$, the results follow from the above Theorem 6.2.4. □

7 Conclusion

The complex numbers encapsulate a much richer structure than the real numbers, yet keeping most of the useful results that we developed in real analysis. Throughout this project, we have seen how elementary topics such as sequences, limits and derivatives are very similar between $\mathbb{R}$ and $\mathbb{C}$. However, there is actually one aspect which is drastically different: integration. Unfortunately, it requires much more machinery and knowledge beyond this project's scope. Interested reader may refer to [22, Chapter 2] for details.

Nevertheless, there are still some insights that can be extracted through this incomplete survey of complex numbers and its applications. This is best seen by in Section 6.1 and Section 6 where we introduce various geometrical concepts. Actually, much more can be done with the developed results. For example, Möbius mapping and stereographic projection introduced in Definition 6.2.2 and Section 6.1 can actually be used to understand another model of geometry called **hyperbolic geometry**. In this model, distance between two points (represented by complex number z_1, z_2) are no longer the usual euclidean distance $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y^2)}$. Instead of translation and rotation, all Möbius mappings now preserve the new notion "distance" between two points. Further information can be found in [1, Chapter 1 to 3].

Also, in Section 4, we see how the notions in topology arises. And in fact, the similar treatment can be done in any other spaces with a notion of distance. That is, given a **metric space**, such as $\mathbb{R}^n$ for any integer n, we can derive most of the result in the section. This is another vital part in analysis, readers are referred to [6] foe more details.

In a nutshell, complex analysis is a very rich subject with various visualization. This is especially relevant when we discuss about mappings and regions. Interested readers can refer to [16] for more diagrams.

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